MAPLIGHT THERAPEUTICS, INC.

(NASDAQ: MPLT)

OUTPERFORM

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Biopharma / Neuroscience

Reason for report:

INITIATION

A New and Improved Cobenfy; Initiating with OP and $30 PT

November 21, 2025

- **Bottom Line: We are initiating coverage of MPLT with an Outperform rating and a SOTP-based $30 PT. Our investment thesis is four-fold:**
  - We are still bullish on the muscarinic space, believing that physicians appreciate the differentiated profile of this new mechanism, that these agents could have broad applicability in treating schizophrenia, AD psychosis, and other CNS disorders (e.g., bipolar disorder, dyskinesias, autism spectrum disorder), and that for schizophrenia the agents seem to benefit the positive, negative and cognitive symptoms.
  - Bristol [BMY, OP, Risinger] / Karuna's Cobenfy demonstrated robust evidence across 3 clinical trials in schizophrenia and Lilly [LLY, OP, Risinger] with Xanomeline (part of Cobenfy) demonstrated positive data in ADP many years ago (and we hope this will be confirmed with the imminent Cobenfy ADP data), so the class is partly de-risked in two key indications.
  - The key asset ML-007C-MA has a potentially better clinical profile than Cobenfy, as it's more potent on both the M1 and M4 receptors, has potentially better dosing (QD vs BID in schizophrenia, BID vs TID in Alzheimer's disease psychosis), shorter titration, no fasting requirement, and better safety because of an improved fixed-dose co-formulation of an M1/M4 agonist with a peripherally acting anticholinergic (PAC) that improves peripheral exposures to better mitigate the cholinergic AEs vs Cobenfy.
  - ML-004 in autism spectrum disorder (ASD) is a wild card and not in the valuation today, but ASD is a high unmet need with nothing approved, and this asset is in Phase 2 development with data coming in 2H26.
- **Our $30 PT is based on a SOTP analysis that forecasts cash flows into the 2040s, with a 10% discount rate.**

Key Stats

S&P 600 Health Care Index:	**3,006.19**
Price:	**$13.29**
Price Target:	$30.00
52-Week High – Low:	$20.86 - $13.28
Shares Outstanding (mil):	43.6
Market Capitalization (mil):	$579.4
Book Value/Share:	$0.00
Cash Per Share:	$11.00
Dividend (ann):	$0.00
Dividend Yield:	0.0%

	Dec Yr		
	2024A	**2025E**	**2026E**
Q1	--	0.0A	--
Q2	--	0.0A	--
Q3	--	0.0A	--
Q4	--	0.0	--
FY Rev	0.0	0.0	0.0
Q1	--	($0.95)A	--
Q2	--	($1.05)A	--
Q3	--	($1.80)A	--
Q4	--	($1.33)	--
FY EPS	($2.96)	($5.13)	($4.30)
P/E	NM	NM	NM

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Source: Company Information and Leerink Partners LLC Research.

- **Key risks to our investment thesis include:**
  - ML-007C-MA fails to demonstrate efficacy in the Phase 2 trial in schizophrenia in 2H26
  - Bristol's Cobenfy fails to demonstrate efficacy in Alzheimer's disease psychosis in 4Q25
  - New safety signal emerges in the schizophrenia or Alzheimer's disease psychosis programs
  - Competing M4 agents (e.g., Neurocrine's NBI-'568) demonstrate a better clinical profile

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Investment Thesis and Company Snapshot

MapLight: Company Overview

MapLight is a clinical-stage biotechnology company focused on developing novel therapies targeting neuropsychiatric indications

Key Facts	
Founded in	2018; Headquarters in California
Listing	MPLT (Nasdaq)
CEO	Chris Kroeger
Employees	> 100 (as of 9/19/2025)
Areas of Focus	Schizophrenia, AD psychosis

- **The key asset ML-007C-MA is a fixed-dose combination of an M1/M4 muscarinic agonist co-formulated with fesoterodine (a peripherally acting anticholinergic) and is in Phase 2 development for the treatment of schizophrenia**
  - Muscarinic agents represent a novel mechanism of action in schizophrenia, and M1/M4 agents have the potential for broad efficacy and improved safety profile with no EPS or metabolic AEs relative to SOC
  - The co-formulation of ML-007C-MA synchronizes the PK of the M1/M4 agonist with fesoterodine to mitigate peripheral cholinergic AEs and improve tolerability over Bristol's Cobenfy (co-formulation of xanomeline and trospium)
  - The Phase 1 trial in healthy adults demonstrated favorable safety with both once- and twice-daily dosing, with 1-day titration and no fasting requirement
- **ML-007C-MA is in Phase 2 development for the treatment of Alzheimer's disease psychosis (ADP)**
  - ADP is a large market with significant unmet need and widespread off-label use of antipsychotics associated with increased risk of mortality
  - The Phase 1 in healthy elderly demonstrated generally a favorable safety with BID dosing (vs TID for Cobenfy)
- **ML-004 is an IR and ER of zolmitriptian in Phase 2 development for the treatment of autism spectrum disorder (ASD)**
- **The company has 2 other preclinical assets: ML-021 (M4 antagonist) and ML-009 (GPR52 PAM)** in development for the treatment of motor deficits in Parkinson's and for the treatment of agitation and aggression, respectively

Timeline of key catalysts

2026E

- **2H26:** Top-line data from the Phase 2 ZEPHYR trial in **schizophrenia**
- **2H26:** Top-line data from the Phase 2 IRIS trial in **ASD**

2027E

- **2H27:** Top-line data from the Phase 2 VISTA trial in **ADP**

IR = Immediate release; ER = Extended release; ASD = Autism spectrum disorder; ADP = Alzheimer's disease psychosis
Sources: Company reports and presentations; Leerink Partners Research

MapLight: Investment Summary

INVESTMENT THESIS:

1. **We are still bullish on the muscarinic space, believing that physicians appreciate the differentiated profile of this new mechanism, that these agents could have broad applicability in treating schizophrenia, AD psychosis, and other CNS disorders (e.g., bipolar disorder, dyskinesias, autism spectrum disorder), and that for schizophrenia the agents seem to benefit the positive, negative and cognitive symptoms**
2. **Bristol/Karuna's Cobenfy demonstrated robust evidence across 3 clinical trials in schizophrenia and Lilly with Xanomeline (part of Cobenfy) demonstrated positive data in ADP many years ago (and we hope this will be confirmed with the imminent Cobenfy ADP data), so the class is partly de-risked in two key indications.**
3. **The key asset ML-007C-MA has a potentially better clinical profile than Cobenfy, as it's more potent on both the M1 and M4 receptors, has potentially better dosing (QD vs BID in schizophrenia, BID vs TID in Alzheimer's disease psychosis), shorter titration, and no fasting requirement, and better safety because of an improved fixed-dose co-formulation of an M1/M4 agonist with a peripherally acting anticholinergic (PAC) that improves peripheral exposures to better mitigate the cholinergic AEs vs Cobenfy**
4. **ML-004 in autism spectrum disorder (ASD) is a wild card and not in the valuation today, but ASD is a high unmet need with nothing approved, and this asset is in Phase 2 development with data coming in 2H26**

VALUATION:

Our $30 PT is based on a SOTP analysis that forecasts cash flows into the 2040s, with a 10% discount rate.

KEY RISKS:

- ML-007C-MA fails to demonstrate efficacy in the Phase 2 trial in schizophrenia in 2H26
- Bristol's Cobenfy fails to demonstrate efficacy in Alzheimer's disease psychosis in 4Q25
- New safety signal emerges in the schizophrenia or Alzheimer's disease psychosis programs
- Competing M4 agents (e.g., Neurocrine's NBI-'568) demonstrate a better clinical profile

Sources: Company reports and presentations; Leerink Partners Research

Financials, Valuation, and Catalysts

MapLight Therapeutics (MPLT): Key Catalysts

Candidate	Indication	Catalyst	Expected Timing
ML-007MA	Schizophrenia	Phase 2 readout	2H26
ML-007MA	AD Psychosis	Phase 2 readout	2H27
ML-004	Autism spectrum disorder	Phase 2 readout	2H26

Sources: Company reports and presentations; Leerink Partners Research

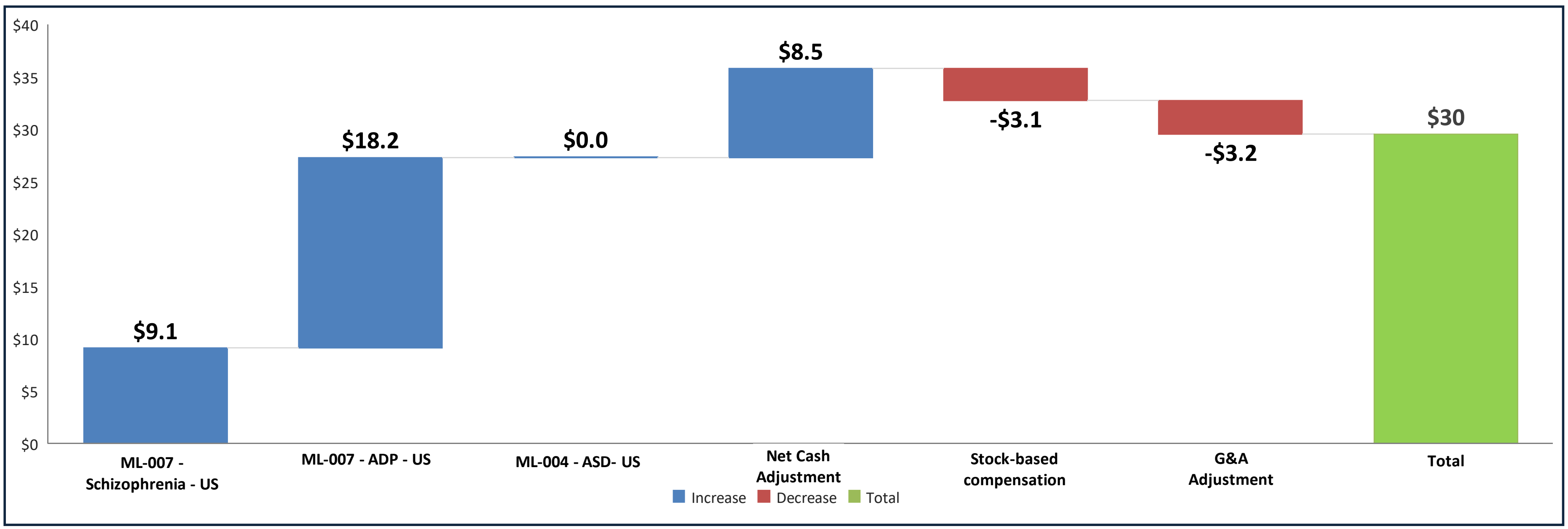
Maplight Therapeutics (MPLT): Sum-of-the-Parts (SOTP) Valuation

MapLight

Product	Non- Risk Adj. Peak Revenue ($M)	Stage of Development	Non-Risk Adj. NPV ($M)	Probability of Success	Risk-Adj. NPV ($M)	Expected Launch	Risk-Adj. Net Current Value / Share ($)
Pipeline							
ML-007 - Schizophrenia - US	$1,015	Phase 2	$715	65%	$464	2030E	$9.1
ML-007 - ADP - US	$1,815	Phase 2	$1,430	65%	$929	2032E	$18.2
ML-004 - ASD- US	$2,005	Phase 2	$3,362	0%	$0	2031E	$0.0
Pipeline Total	$4,835	---	$5,507	---	$1,394		$27.3
Adjustments							
Net Cash	---	---	$433	---	$433	---	$8.5
Stock-based compensation			-$157		-$157		-$3.1
G&A	---	---	-$163	---	-$163	---	-$3.2
			Non-Risk Adj.		Risk-Adj.		Risk-Adj.
Total Asset Value	---	---	$5,776	---	$1,663	---	$30
Share Count (M) - Fully Diluted (4Q25)	51						
Total Risk-Adjusted Equity Value*	$30						

**We exclude unallocated R&D to account for corporate technology franchise value; However, this valuation for our price target includes net cash and G&A adjustments.*

SOTP Valuation Breakdown by Product and Region



Sources: Company reports and presentations; Leerink Partners Research

Maplight Therapeutics (MPLT): Upside/Downside Scenarios

Key Variable:	Downside	Current stock level[1]	OUTPERFORM Our view	Blue Sky Scenario
	Price: $4 -70%	Stock price: $14	PT: $30 +115%	Price: $45 +220%

Key SOTP Valuation Details

ML-007C-MA – Schizophrenia – US	▪ Peak sales of ~$700M, 45% POS	▪ Peak sales of ~$1B, 45% POS	▪ Peak sales of ~$1B, 65% POS	▪ Peak sales of ~$1B, 100% POS
ML-007C-MA – ADP – US	▪ 0% POS	▪ Peak sales of ~1.7B, 20% POS	▪ Peak sales of ~$1.7B, 65% POS	▪ Peak sales of ~$1.7B, 100% POS
ML-004 – ASD – US	▪ 0% POS	▪ 0% POS	▪ 0% POS	▪ 0% POS

(1) Stock at close on 11/20/2025
(2) Blue Sky Scenario is just for lead asset

POS = Probability of success

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Sources: Company reports and presentations; Leerink Partners Research

MapLight Therapeutics (MPLT): Overview of Key Debates

Key Debates	Our View	Risks
1 **Can ML-007C-MA differentiate from Cobenfy by improving the safety profile or ease-of-use in SCZ?**	**We think there is a good chance for ML-007C-MA to improve the side effect burden over Cobenfy and achieve a simpler dosing regimen.** 1. The company has optimized ML-007 and the peripherally acting anticholinergic (PAC) dosing such that the shapes of the PK curves match peripheral agonism and antagonism while maintaining central activity; as a result, we believe both anti- and pro-cholinergic side effects may be mitigated, improving tolerability 2. We believe early safety data from Phase 1 studies support the viability of this strategy, where GI AEs (such as vomiting and diarrhea) were notably absent 3. ML-007C-MA may offer a meaningfully more convenient dosing profile than Cobenfy, with no fasting requirement, the potential for once-daily dosing, and a faster titration schedule (1 day vs 3-8 days) 4. Growing enthusiasm around the muscarinic class lowers the barrier to entry for a fast-follower, as the therapeutic story is already familiar to physicians	▪ Nausea rates are high in the Phase 2 trial ▪ The BID dosing arm performs better than the QD arm
2 **Will the muscarinic class be effective in treating ADP?**	**We like the opportunity for ML-007C-MA in ADP given the prior positive data in treating psychosis with xanomeline and the potential for positive read-through from the Phase 2 trial of Cobenfy in ADP** 1. Prior work demonstrated that xanomeline prevented delusions, hallucinations, and other psychotic symptoms in AD patients and demonstrated a positive effect on cognition after 6 months on therapy 2. The Phase 2 ADP trial with Cobenfy could meaningfully de-risk the muscarinic mechanism in this population, increasing the POS for ML-007C-MA 3. There remains a substantial unmet need in ADP, where psychosis is frequently managed with off-label atypical antipsychotics despite their association with increased mortality in elderly patients	▪ Cobenfy shows weak efficacy in the Phase 2 trial in ADP ▪ New safety signal arises in larger trials in elderly populations
3 **Is there evidence to support a role for 5HT1B agonists to treat autism?**	**We view the data linking serotonergic dysfunction to autism as intriguing and potentially supportive of 5-HT1B agonism as a viable therapeutic approach** 1. Serotonergic abnormalities are common among ASD patients 2. Preclinical data demonstrates that 5HT1B receptor activation with ML-004 can modulate both sociability and aggression	▪ Preclinical models for psychiatric conditions don't often predict efficacy in humans

ADP = Alzheimer's disease psychosis

Sources: Company reports and presentations; Leerink Partners Research

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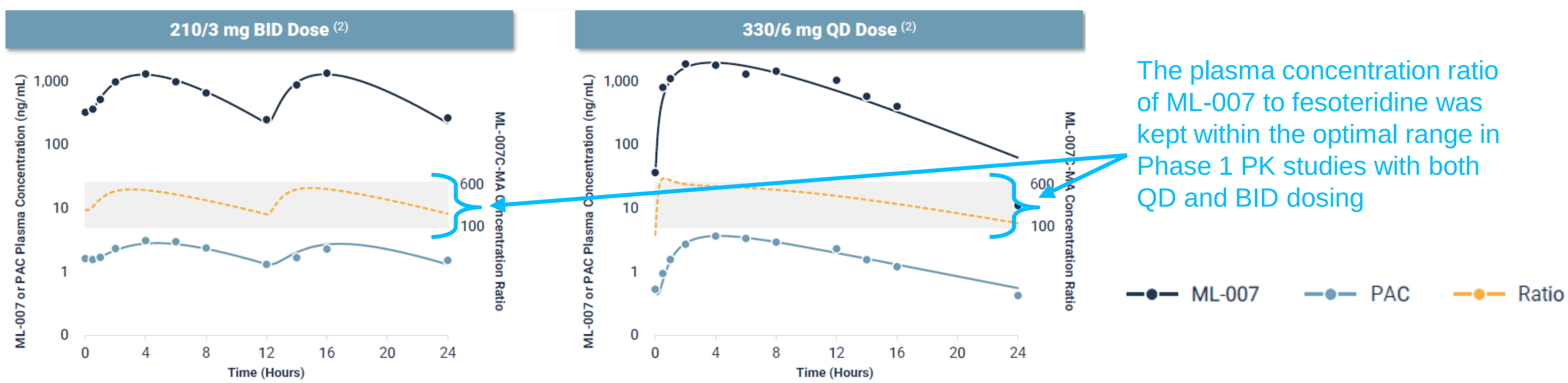
ML-007C-MA in Schizophrenia

ML-007C-MA is a combination therapy of an M1/M4 agonist and fesoteridine (peripherally acting anticholinergic) that has potential to improve tolerability over Cobenfy while maintaining efficacy

What's the hook?	▪ ML-007C-MA is a coformulation of the M1/M4 agonist ML-007 and fesoteridine that optimizes the PK of both drugs to match agonism and antagonism in the periphery without sacrificing strong activation in the brain ▪ This optimized PK synchronization has the potential to improve upon the safety profile seen with Cobenfy and increase tolerability ▪ The therapeutic potential of the muscarinic class in schizophrenia has been largely de-risked following the successful Phase 3 trials of Cobenfy and subsequent FDA approval ▪ Additional differentiation includes faster titration and no fasting requirement
MoA	▪ ML-007 binds to M1 and M4 muscarinic receptors in the brain, which decrease dopaminergic tone in the striatum, leading to antipsychotic effects ▪ Fesoteridine is peripherally restricted and blocks muscarinic activation outside of the brain ▪ Co-administration of ML-007 and fesoteridine produces central M1/M4 agonism while minimizing peripheral effects through opposing mechanisms of action

Data Highlights

Phase 1 data showing synchronized PK between ML-007 and fesoteridine



Development status	▪ A Phase 2 trial in schizophrenia is ongoing with data expected in 2H26 ▪ A Phase 2 trial in AD psychosis is ongoing with data expected in 2H27
Patents	▪ MapLight owns three patents in the US for ML-007 covering composition of matter and methods of use in schizophrenia and ADP, which are expected to expire between 2031 and 2032, excluding of patent term extensions ▪ MapLight also holds pending patent applications in the US and internationally for the combination of ML-007 and a PAC to treat schizophrenia and ADP, which are expected to expire in 2042 excluding patent term extensions

Sources: Company reports and publications; Leerink Partners Research
PAC = peripheral anticholinergic

Key muscarinic assets approved or in development

Asset	Company	MoA	Indication	Stage	Product Hook	Next Catalyst	Timing
Cobenfy	BMY	M1/M4 preferring agonist	Schizophrenia AD psychosis	Approved	First-in-class muscarinic antipsychotic	Top-line data from first Phase 3 in ADP	4Q25
ML-007C-MA	MapLight	M1/M4 preferring agonist	Schizophrenia AD psychosis	Phase 2	Potential for improved safety and ease-of-use over Cobenfy	Top-line data from Phase 2 in schizophrenia/ADP	2H26/ 2H27
Emraclidine	ie / Cerevel (, OP, Risinger)	M4 PAM	AD psychosis PD psychosis Schizophrenia	Phase 2	Potential for improved safety profile by avoiding M1	Phase 2 study start in ADP / PDP / LBD and adjunctive schizophrenia	2026
NBI-'568	Neurocrine (NBIX, OP)	M4 selective agonist	Schizophrenia Bipolar mania	Phase 3	Better safety than Cobenfy so far	▪ Phase 2 study start in bipolar mania ▪ First Phase 3 data readout in schizophrenia	4Q25 / 2027 for bipolar mania / SCZ
NBI-'570	Neurocrine	M1/M4 selective agonist	Psychosis	Phase 1	Selective for M1 or M4 only; No use of a PAC	Phase 1 data readout	4Q25
NBI-'569	Neurocrine	M4 preferring agonist (M1/M4 selective agonist)	Neuropsychiatry	Phase 1	The goal is to assess M4 biased contribution to the overall efficacy; No use of a PAC	Phase 1 data readout	4Q25
NBI'567	Neurocrine	M1 preferring agonist (M1/M4 selective agonist)	Neuropsychiatry	Phase 1	The goal is to assess M1 biased contribution to the overall efficacy; No use of a PAC	Phase 1 data readout	4Q25
NMRA-861/ Next-gen	Neumora (NMRA, Not Rated)	M4 PAM	Schizophrenia	Phase 1	Highly potent with potential best-in-class pharmacology	SAD/MAD and PK data readout	1Q26
NTX-253	Neurosterix (Private)	M4 PAM	Psychosis Mood disorders	Phase 1	Potential for wider therapeutic index than that of emraclidine	FIH trials stars	4Q25
TerXT	Terran Biosciences (Private)	M1/M4 agonist	Schizophrenia	Phase 1	Long-acting injections with multi-month duration, 505(b)(2) pathway for approval	The Phase 1 data demonstrated favorable safety and tolerability profile (LINK)	

Sources: Company reports and presentations; BiomedTracker, Leerink Partners Research

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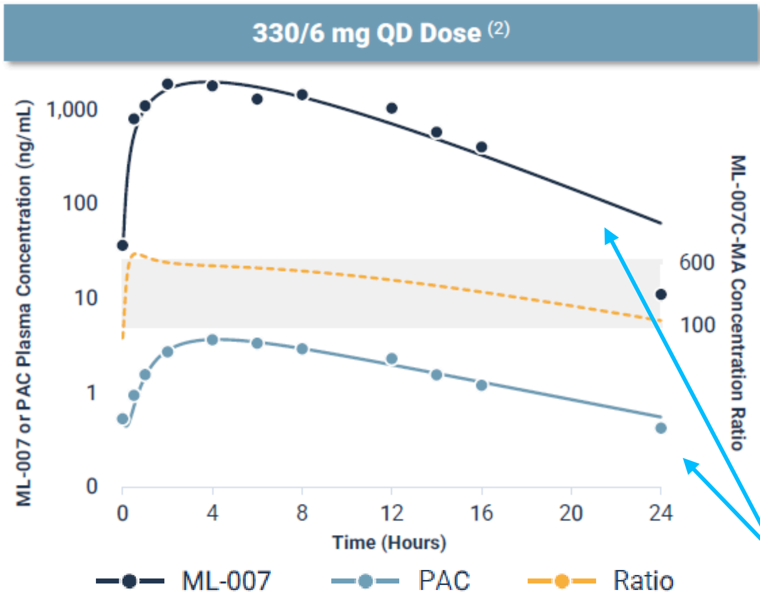
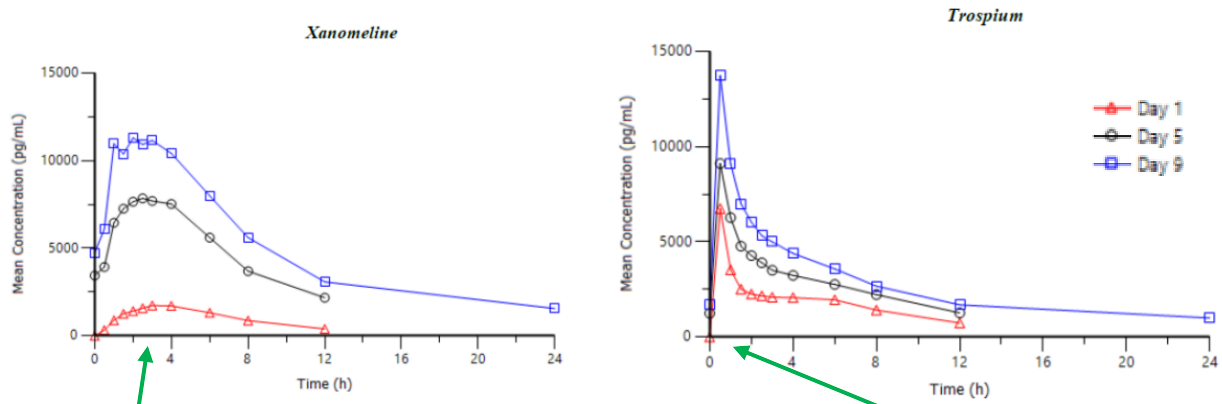
ML-007C-MA is designed to match the PK for the M1/M4 agonist component (ML-007) with the peripheral anti-cholinergic (PAC) component (fesoterodine) to mitigate peripheral cholinergic AEs

PK profile of ML-007 and fesoterodine



PK profile of xanomeline and trospium



Xanomeline peaks after 2-3 hours and has a long tail, whereas Trospium peaks within the first hour and declines rapidly

The shapes of the ML-007 and fesoterodine curves are well-matched

MapLight optimized the dosing of ML-007 and fesoterodine in four Phase 1 studies to maintain a relatively consistent serum concentration ratio within a range that balances pro- and anti- cholinergic effects

- Importantly, the PK curves with ML-007 and fesoterodine are better matched than xanomeline and trospium, indicating a consistent balance between pro- and anti-cholinergic activity over the dosing period
- The synchronization is important because insufficient antagonist activity to neutralize agonist activity results in procholinergic side effects, whereas excessive antagonist activity results in anticholinergic side effects

Additionally, the PK curves suggest dosing may be once daily compared to Cobenfy's BID dosing

Key takeaway:

- **We believe that the matching of the exposures between the M1/M4 and the PAC that MapLight uses compares favorably to that of Cobenfy and should be a helpful differentiator and drive a better AE profile**

Sources: Company reports and publications; Leerink Partners Research

ML-007C-MA demonstrated an improved safety and tolerability profile compared to Cobenfy with no episodes of anti-cholinergic safety signal in the Phase 1 trial in healthy adults

Summary of safety data in Phase 1

Both BID and QD doses are being tested in Phase 2

	Placebo Combined (N = 8) [2]	210/3 mg BID (N = 6)	330/6 mg QD (N = 6) [3]
Subjects With Any TEAE [1]	2 (25%)	3 (50%)	4 (67%)
Mild	2 (25%)	3 (50%)	4 (67%)
Moderate [3]	0 (0%)	1 (17%)	2 (33%)
Severe	0	0	0
Most Common TEAEs (>1 Subject in any Dose Group)			
Chills	0	0	2 (33%)
Hyperhidrosis	0	1 (17%)	2 (33%)
Nausea	0	3 (50%)	3 (50%)
Dizziness	0	3 (50%)	2 (33%)
Dyspepsia	0	0	2 (33%)

Pro-cholinergic AEs

Importantly, there were no anti-cholinergic effects

Key takeaways:

- **While we believe that the pro-cholinergic AEs are still noticeable (although in a small N), especially for nausea at 50%, there were still fewer cases overall relative to Cobenfy**
- **The lack of anti-cholinergic AEs is a benefit**

Sources: Company reports and publications; Leerink Partners Research

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Cobenfy shows a safety profile that is tolerable, but patients experienced several anti- and pro-cholinergic AEs that were consistent between the Phase 1 and Phase 3 trials

Safety data from Cobenfy Phase 1 trial

AEs	Xanomeline + Placebo (n=33)	KarXT (n=35)
Any Cholinergic AEs (p=0.016)	**64%**	**34%**
Nausea	24%	17%
Vomiting	15%	6%
Diarrhea	21%	6%
Sweating	49%	20%
Salivation	36%	26%

- Of the top 5 AEs seen in the Phase 3 trial with Cobenfy, only nausea and dyspepsia emerged in the Phase 1 trial with ML-007C-MA
- Vomiting was a consistent AE between the Phase 1 (6%) and Phase 3 (16%) with Cobenfy, but it did not emerge with ML-007C-MA

Safety data from Cobenfy Phase 3 trial

Variable	Participants, No. (%) Xanomeline-trospium chloride (n = 125)	Placebo (n = 128)
Any TEAE	88 (70.4)	64 (50.0)
Serious TEAE[a]	1 (0.8)	0
TEAE leading to discontinuation	8 (6.4)	7 (5.5)
TEAE occurring in ≥5% of participants in the xanomeline-trospium chloride group		
Nausea	24 (19.2)	2 (1.6)
Dyspepsia	20 (16.0)	2 (1.6)
Vomiting	20 (16.0)	1 (0.8)
Constipation	16 (12.8)	5 (3.9)
Headache	14 (11.2)	15 (11.7)
Hypertension[b]	8 (6.4)	2 (1.6)
Insomnia	7 (5.6)	10 (7.8)
Diarrhea	7 (5.6)	1 (0.8)

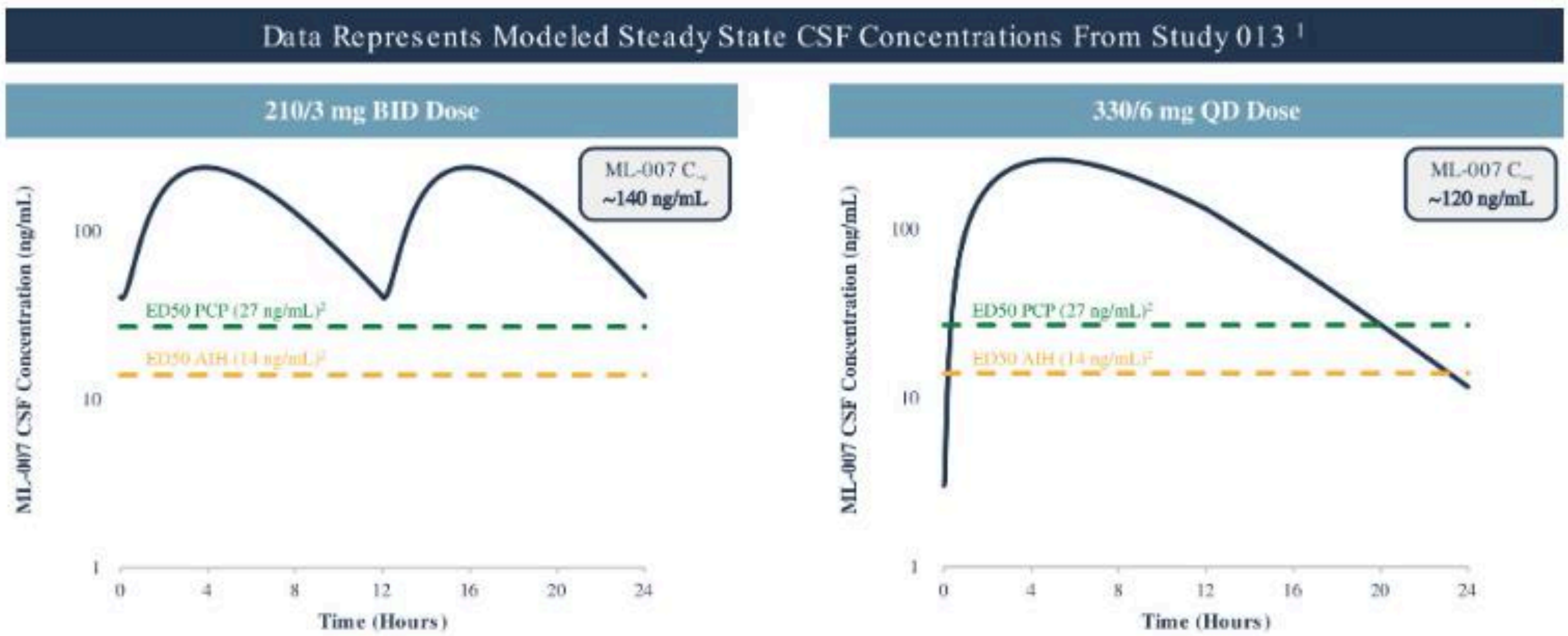
Key takeaways:

- **Early clinical data indicate that ML-007C-MA may offer an improved GI side-effect profile versus Cobenfy, although we are watching the rates of nausea**

Sources: Company reports and publications; Leerink Partners Research, Kaul et al 2024

The PK profile of ML-007C-MA supports the possibility for either QD or BID dosing, whereas Cobenfy can only be dosed BID in schizophrenia due to tolerability

Modeled CSF Exposures at the Phase 2 target doses



1. ML-007 lines represents modeled estimates for CSF exposure based on CSF sampling conducted during Study 013.
2. ML-007 was tested in multiple animal models. CSF exposures were correlated with pharmacodynamic effects to define a target efficacious range of 14 to 27 ng/mL, based on the AIH and PCP models, respectively.

- **The target doses of 210/3 mg BID and 330/6 mg QD resulted in predicted CSF exposures at or above the target range**
- **Importantly, there were no severe AEs with the QD ML-007C-MA dose in Phase 1 trials, and we are watching for continued tolerability in Phase 2 with a larger cohort**
  - Note that in a Phase 1 study of Cobenfy dosed QD (healthy elderly patients), less than half of the participants were able to tolerate therapy

Key takeaway:

We believe that ML-007C-MA has the potential to differentiate with the dosing frequency compared to Cobenfy, and we are watching for continued tolerability with the QD dosing arm in the ongoing Phase 2 trial

Sources: Company reports and publications; Leerink Partners Research

Due to food effects, Cobenfy is dosed in the fasted state, whereas ML-007C-MA shows higher exposures and better tolerability in the fed state which has the potential to improve ease-of-use

Cobenfy food effects

Effect of food: PK in fed state (compared to fasted state)

High fat meal[b]	C_{max}	Unchanged	Reduced 70% to 75%
	AUC	Increased 30%	Reduced 85% to 90%
Low fat meal[b]	C_{max}	Unchanged	Reduced 70% to 75%
	AUC	Unchanged	Reduced 85% to 90%

(Continued)

- Both high and low-fat meals reduced Trospium exposures >70%,
- As a result, dosing in the fasted state is included on the label

ML-007C-MA food effects

- ML-007 PK exposures were higher in the fed conditions compared with that in the fasted condition, and there was no food effect on the fesoterodine PK
- There was no difference in exposures between low- and high- calorie meals
- ML-007C-MA was well tolerated in fed and fasted conditions, but tolerability was improved when administered in the fed conditions
- As a result, participants are dosed proximal to a meal in the Phase 2 study

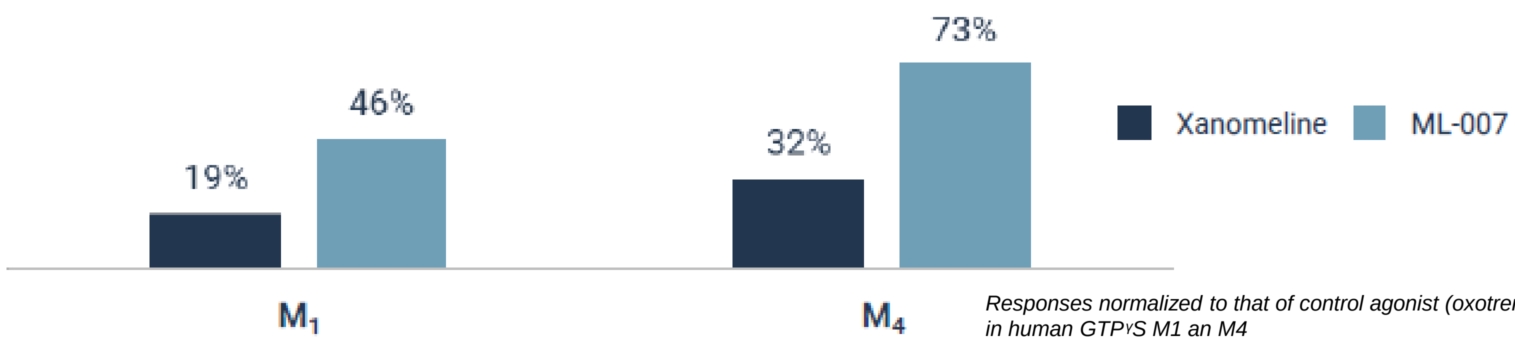
Key takeaways:

- **ML-007C-MA does not require fasting before dosing, which could lead to a dosing advantage vs Cobenfy**

Sources: Company reports and publications; Leerink Partners Research

ML-007 (M1/M4 agonist component) has stronger relative potency at both M1 and M4 receptors in *in vivo* models compared to xanomeline (M1/M4 agonist component in Cobenfy)

In Vitro assay show stronger relative peak intrinsic activity



Responses normalized to that of control agonist (oxotremorine) in human GTPγS M1 an M4

ML-007 has greater bioavailability given the low protein binding

	Xanomeline	ML-007
Protein Binding	▲	▼
Oral Bioavailability	▼	▲
Peripheral Tissue Accumulation	Extensive (incl. liver)	Low

- Xanomeline and Trospium have 95% and 80% plasma protein binding, respectively

Key takeaways:

- **The strong potency of ML-007 at M1 and M4 receptors and the favorable bioavailability drives confidence that ML-007C-MA should be able to match the solid efficacy of Cobenfy in schizophrenia, at a minimum**

Sources: Company reports and publications; Leerink Partners Research

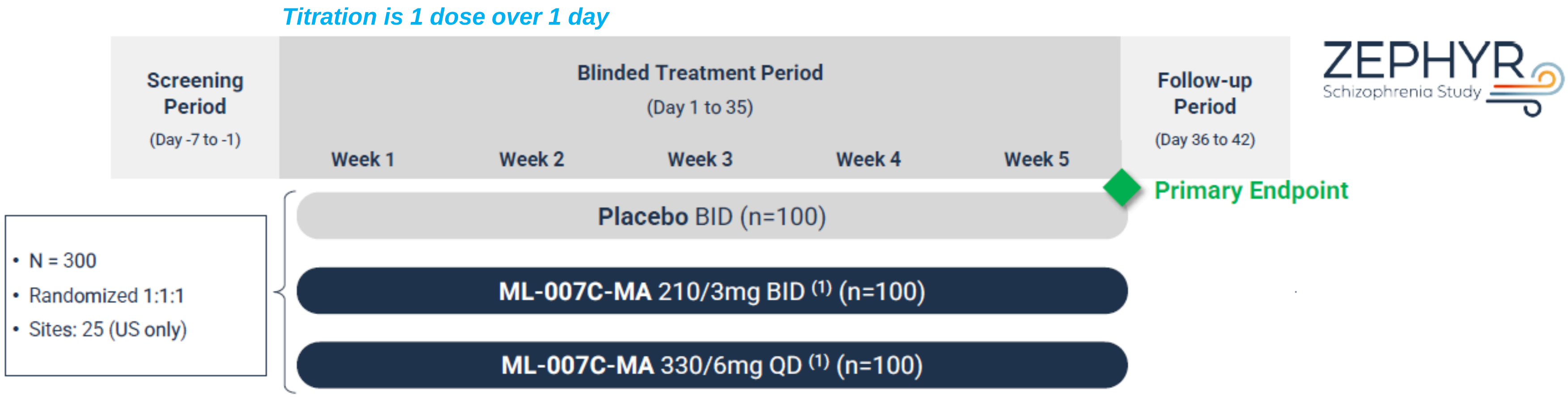
ML-007C-MA differentiation summary

		Cobenfy (BMY)	ML-007C-MA (MPLT)	Favors ML-007C-MA
PK / PD	Activity at M1	19%	46%	✓
	Activity at M4	32%	73%	✓
	Bioavailability	Lower	Higher	✓
	Protein binding	Higher	Lower	✓
Titration length	Schizophrenia	3-8 days	1 day	✓
	ADP	5 weeks	1 week	✓
Dosing Frequency	Schizophrenia	BID	QD or BID	
	ADP	TID	BID	✓
Food effect	Fasting requirement	Yes	No	✓
Safety: Pro-cholinergic	Nausea	Ph1: 17%; Ph3: 19.2%	QD: 50%; BID: 50%	
	Vomiting	Ph1: 6%; Ph3: 16%	None reported	✓
	Dizziness	None reported	QD: 50%; BID: 33%	
	Hyperhidrosis	Ph1: 20%; Ph3: 1.8%	QD: 33%; BID: 17%	
	Gastroesophageal reflux disease	Ph3: 0.8%	None reported	✓
Safety: Anti-cholinergic	Dyspepsia	Ph3: 15%	QD: 33%; BID: 0%	
	Constipation	Ph3: 17%	None reported	✓
	Dry mouth	Ph3: 5%	None reported	✓
	Urinary retention	Ph3: 0.6%	None reported	

Note that safety data for Cobenfy is pooled across the Phase 3 program in schizophrenia patients (N =251), while the safety data for ML-007C-MA is from the Phase 1 in healthy adults

Sources: Company reports and publications; Leerink Partners Research

The company is running a Phase 2 trial in schizophrenia with data expected in 2H26



Study Population	• 18 to 64 years of age with diagnosis of schizophrenia with MINI • PANSS 80-120 at screening and baseline • Score ≥4 for two or more positive symptom items • CGI-S score ≥4 • Untreated or recent wash-out of antipsychotics	**Primary Endpoint**	• CFB PANSS at Week 5
		Key Secondary Endpoints	• CFB CGI-S at Week 5 • CFB PANSS positive Marder factor at Week 5 • CFB PANSS negative Marder factor at Week 5

- The company expects top-line data readout in 2H26

Key takeaway:

- **We believe the expectation is that ML-007C-MA demonstrates solid efficacy comparable to that demonstrated by Cobenfy**

Sources: Company reports and publications; Leerink Partners Research

Cobenfy demonstrated robust antipsychotic efficacy in both positive and negative symptoms across different clinical trials, which is the key driver behind the excitement around the muscarinic class

Additionally, Cobenfy demonstrated efficacy in cognition, as an exploratory endpoint in the clinical trials

	Positive	Negative*	Total PANSS	Cognitive*
Emergent-1	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS positive subscale of 5.6 points compared to 2.4 points for placebo **(delta = -3.2; p<0.001)** in EMERGENT-1 (n=182)	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS negative subscale of 3.2 points compared to 0.9 points for placebo **(delta = -2.3; p=0.0055)** in EMERGENT-1 (n=182)	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS total scale of 17.4 points compared to 5.9 points for placebo **(delta = -11.6; p<0.0001)** in EMERGENT-1 (n=182)	There was a signal of efficacy in cognition in the total patient population but only on the composite score in subjects with the highest cognitive impairment
Emergent -2	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS positive subscale of 6.8 points compared to 3.9 points for placebo **(delta = -2.9; p<0.0001)** in EMERGENT-2 (n=256)	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS negative subscale of 3.4 points compared to 1.6 points for placebo **(delta = -1.8; p=0.0055)** in EMERGENT-2 (n=256)	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS total scale of 21.2 points compared to 11.6 points for placebo **(delta = -9.6; p<0.0001)** in EMERGENT-2 (n=252)	KarXT demonstrated significant improvements in the CANTAB composite score compared to placebo at week 5 **in patients with cognitive impairment** at baseline **(d = 0.52; p<0.01)** in a pooled analysis across EMERGENT-2 and EMERGENT-3 trials (n=134)
Emergent -3	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS positive subscale of 7.1 points compared to 3.6 points for placebo **(delta = -3.5; p<0.0001)** in EMERGENT-3 (n=256)	KarXT demonstrated a clinically meaningful reduction on the PANSS negative subscale of 2.7 points compared to 1.8 points for placebo **(delta = -0.9; p=0.12)** in EMERGENT-3 (n=256)	KarXT demonstrated a significant and clinically meaningful reduction on the PANSS total scale of 20.6 points compared to 12.2 points for placebo **(delta = -8.4; p<0.0001)** in EMERGENT-3 (n=256)	

* *Acute negative symptoms*

Key takeaway:

- **Cobenfy as the first to market sets a high bar for efficacy for the rest of the muscarinics in development**

Sources: Company reports and publications; Leerink Partners Research

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Cobenfy's efficacy in schizophrenia is comparable to robust agents and compares favorably to recently approved antipsychotics

Asset	Study (N)	Treatment	Dose/Arms	BL PANSS	Change from BL, PANSS	Difference vs Placebo	Effect size
Latuda (Sunovion, part of Sumitomo [SMPA, not rated])	Study 1 (N=473)	6 weeks	40mg	96.6	-25.7	-9.7	0.36
			120mg	97.9	-23.6	-7.5	
			Olanzapine 15mg	96.3	-28.7	-12.6	
			PBO	95.8	-10.7	---	
Vraylar (ie])	Study 2 (N=489)	6 weeks	40mg	96.5	-19.2	-2.1	0.34
			80mg	96	-23.4	-6.4	
			120mg	96	-20.5	-3.5	
			PBO	96.8	-17	---	
	Study 1 (N = 711)	6 weeks	1.5mg	97.1	-19.4	-7.6	
			3mg	97.2	-20.7	-8.8	
			4.5mg	96.7	-22.3	-10.4	
			PBO	97.3	-11.8	---	
	Study 2 (N = 604)	6 weeks	3mg	96.1	-20.2	-6	
			6mg	95.7	-23	-8.8	
			PBO	96.5	-14.3	---	
	Study 3 (N = 439)	6 weeks	3-6mg	96.3	-22.8	-6.8	
			6-9mg	96.3	-25.9	-9.9	
			PBO	96.6	-16	---	
Rexulti (Otsuka [OTSKF, not rated]/ Lundbeck [HLUNB, MP])	Study 1 (N=536)	6 weeks	2mg	95.9	-20.7	-8.7	0.26
			4mg	94.7	-19.7	-7.6	
			PBO	95.7	-12	---	
	Study 2 (N=540)	6 weeks	2mg	96.3	-16.6	-3.1	
			4mg	95	-20	-6.5	
			PBO	94.6	-13.5	---	
Lybalvi (Alkermes [ALKS, MP])	Study 1 (N=395)	4 weeks	Lybalvi	101.8	-23.9	-6.4	0.56
			Olanzapine 15mg	100.6	-22.8	-5.3	
			PBO	102.7	-17.5	---	
Caplyta (Inta-Cellular, part of J&J [JNJ, MP, Risinger])	Study 1 (N=335)	4 weeks	42mg	88.1	-13.2	-5.8	~0.3
			PBO	86.3	-7.4	---	
	Study 2 (N=450)	4 weeks	42mg	90	-14.5	-4.2	
			PBO	89	-10.3	---	
Cobenfy (BMY)	Pooled EMERGENT data	5 weeks	Cobenfy	97.5	-19.4	-9.9	0.56
			PBO	97	-9.6	---	

Strong effect size for Cobenfy similar to:

- Clozapine: 0.89
- Olanzapine: 0.56
- Risperidone: 0.55
- Paliperidone: 0.49
- Aripiprazole: 0.41

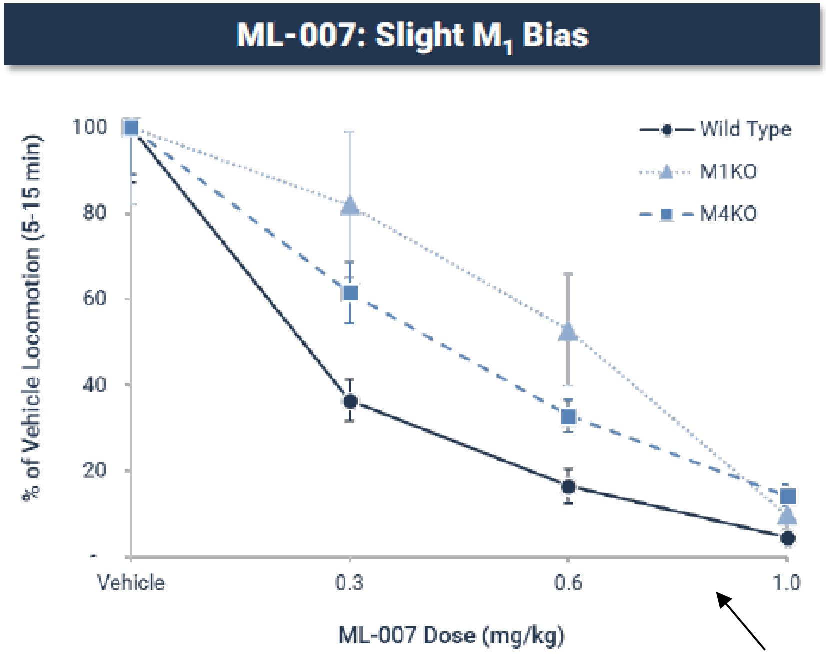
Note that efficacy data for Cobenfy are pooled across all 3 trials in the EMERGENT program

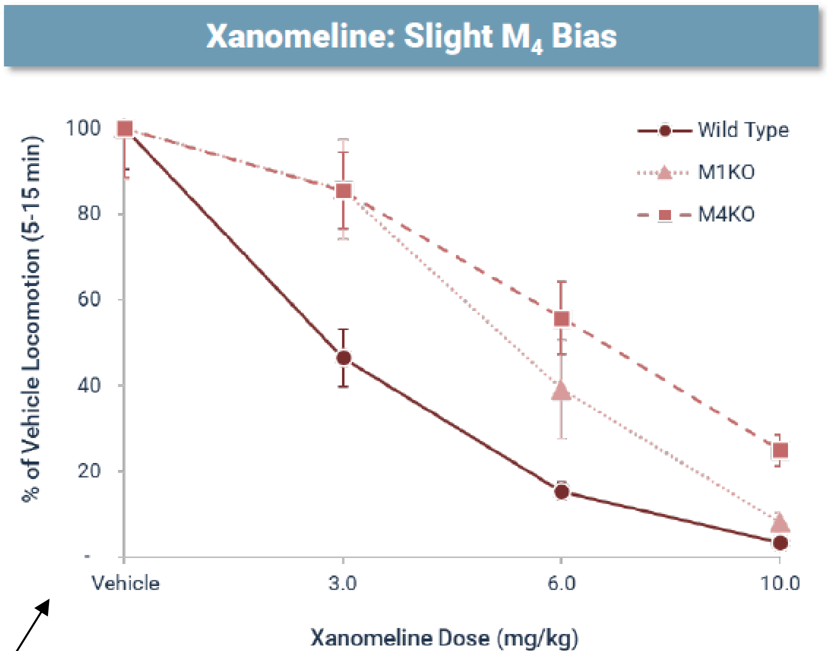
BL = Baseline; PANSS = Positive and Negative Syndrome Scale

Source: Companies reports and presentations; Leerink Partners Research

MapLight provided preclinical data suggesting that both M1 and M4 receptors play a role in the reduction of psychosis in animal models of schizophrenia

Schizophrenia animal model of amphetamine-induced hyperlocomotion





Those findings indicate that the activation of M1 or M4 receptors alone <u>at clinically relevant doses</u> would result in **suboptimal activity, and strong activity at both receptors is associated with robust efficacy on psychosis**

Key takeaways:

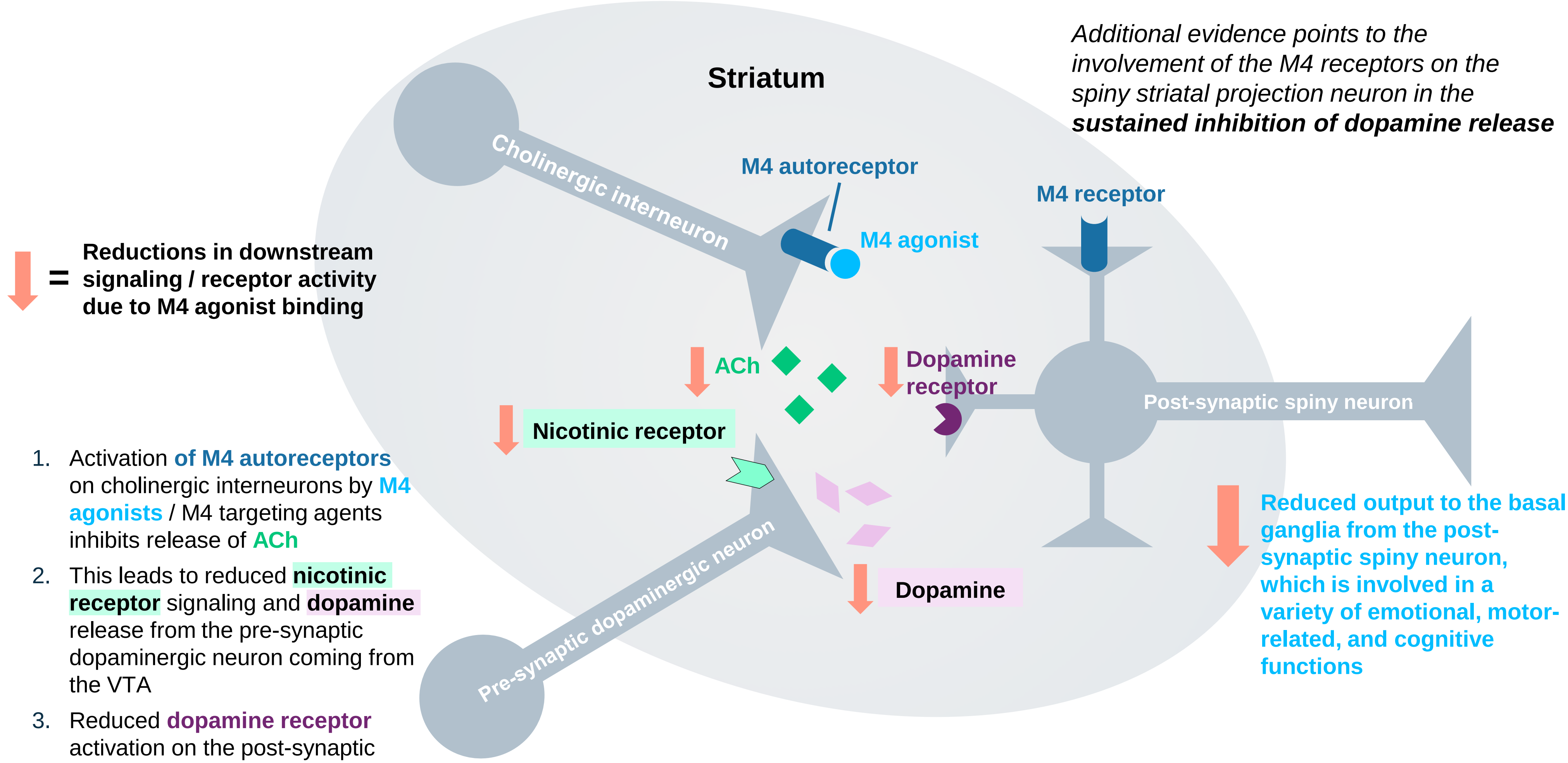
- **The M1 works and the M4 works, but the best results are when both M1 and M4 are activated to match the wild type profile**

Sources: Company reports and publications; Leerink Partners Research

M4 receptor activation makes sense as a target for treating psychosis, since its activation reduces the dopaminergic neurotransmission in the dorsal striatum, leading to alleviation of symptoms of psychosis

M4 agonists indirectly reduce dopaminergic tone in the dorsal striatum by reducing ACh release

Additional evidence points to the involvement of the M4 receptors on the spiny striatal projection neuron in the ***sustained inhibition of dopamine release***



1. Activation **of M4 autoreceptors** on cholinergic interneurons by **M4 agonists** / M4 targeting agents inhibits release of **ACh**
2. This leads to reduced **nicotinic receptor** signaling and **dopamine** release from the pre-synaptic dopaminergic neuron coming from the VTA
3. Reduced **dopamine receptor** activation on the post-synaptic spiny neuron results in reduced output to the basal ganglia

Note: Overactivity of striatal dopamine system is believed to underlie the psychotic symptoms in schizophrenia

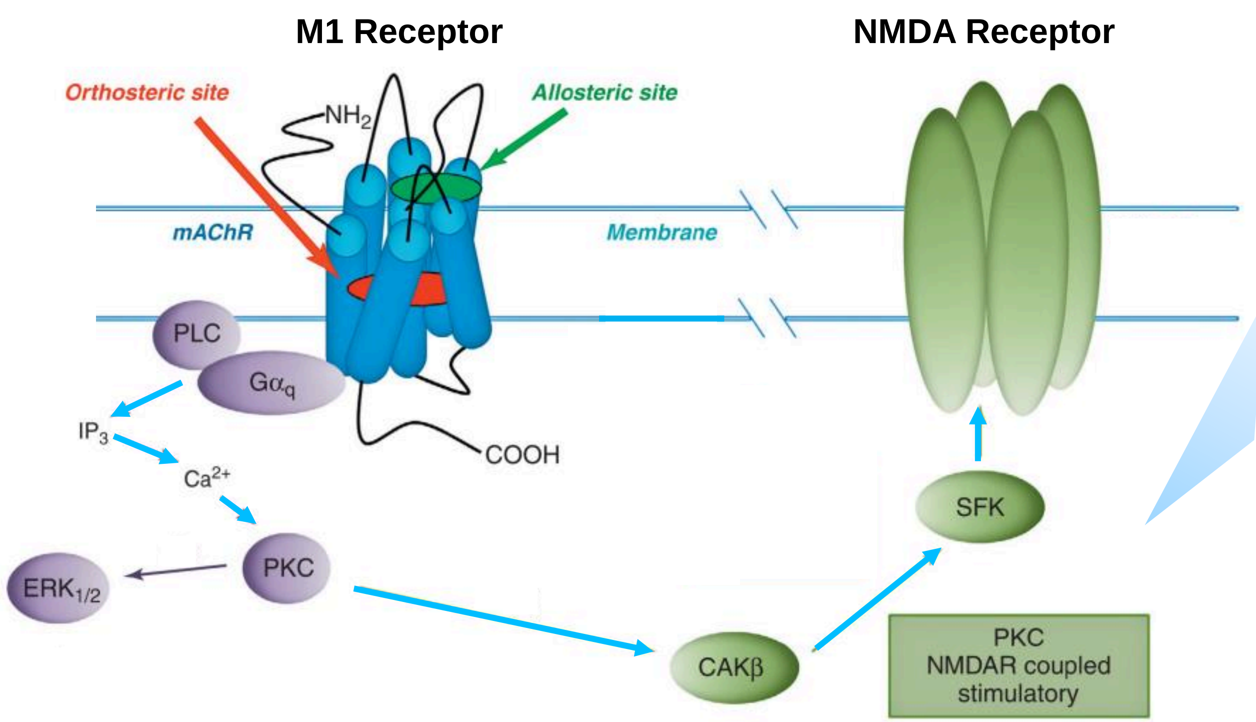
Key takeaway:

- **The reduction of the dopaminergic tone in the striatum via M4 receptor activation appears to lead to the alleviation of the symptoms of psychosis without any blockade of striatal dopaminergic receptors responsible for unwanted side effects of extrapyramidal symptoms and akathisia**

Sources: Foster, Neuron, 2016; Leerink Partners Research
ACh = Acetylcholine; VTA = Ventral tegmental area

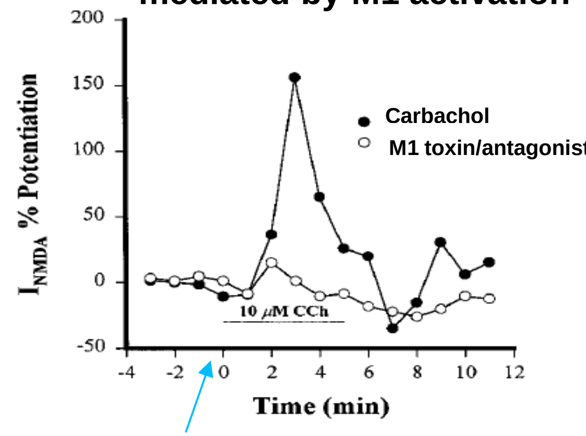
M1 activation was shown to potentiate NMDA-receptor signaling in the hippocampus, which may underlie its potential in improving cognition

M1 activation appears to contribute a pro-cognitive effect and drive an effect on negative symptoms



It has been suggested that the potentiation of the NMDA evoked currents, upon activation of the M1 receptor, is mediated in a protein kinase (PKC)-dependent manner

NMDA potentiation is mediated by M1 activation



Blocking of M1 receptor eliminates the potentiation of NMDA currents induced by carbachol

- Carbachol, a cholinergic (M1 and M3) agonist, was shown to induce rapid potentiation of NMDA currents
- The effect on NMDA current was abolished with an M1 toxin/M1 antagonist, which helps to demonstrate that the NMDA potentiation was caused by M1 receptor activation

Key takeaway:

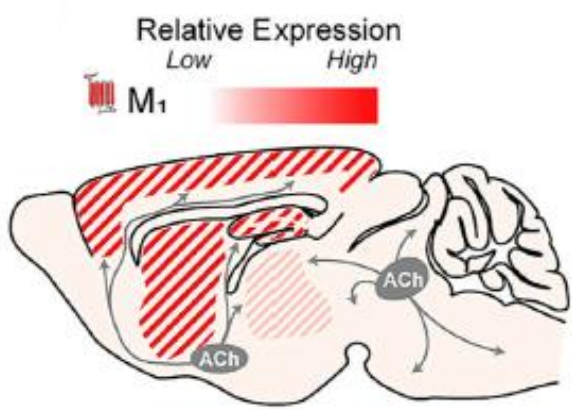
- **NMDA hypofunction has been implicated in the pathophysiology of schizophrenia; since M1 activation was shown to potentiate NMDA currents, it offers new hope as a novel therapeutic target to address the cognitive impairment in schizophrenia**

Sources: Marino, Proc Natl Acad Sci, 1998; Melancon, Drug Discov Today, 2013; Leerink Partners Research

M1 receptor expression in the brain in addition to preclinical data support the pro-cognitive effects of M1 activators (either agonists/PAMs)

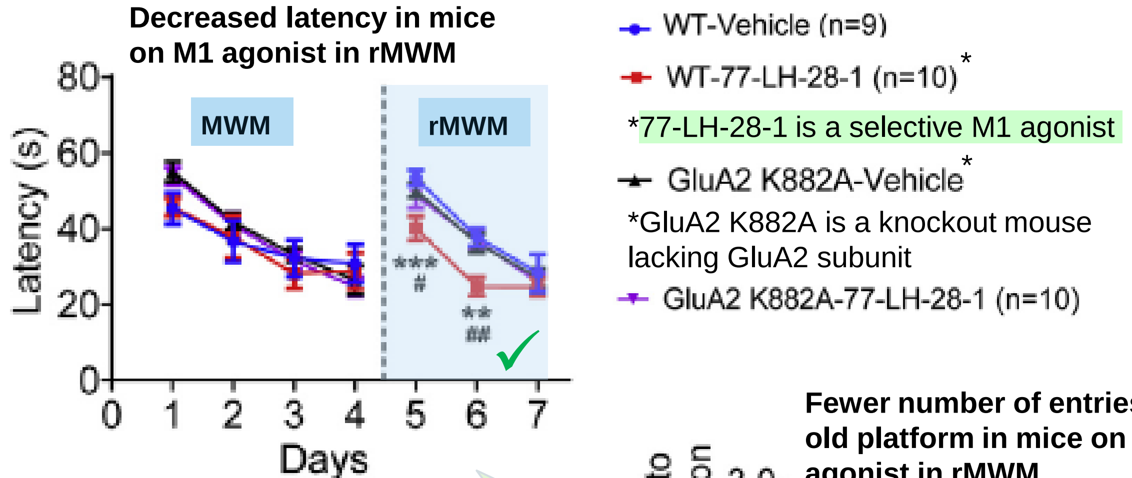
M1 activation facilitates extinction of old memory and reversal learning in mice

Localization of M1 receptors implicates their role in cognition

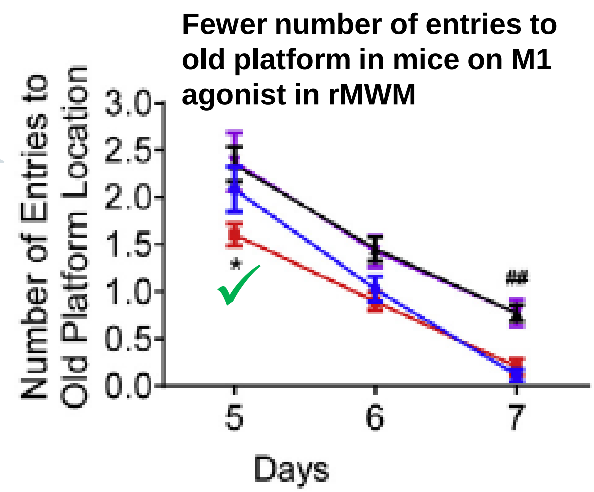


M1 receptors are expressed in the prefrontal cortex, striatum and hippocampus, implicating their role in cognition, movement and memory

Enhanced spatial reversal learning and memory switching in mice on M1 agonists



Based on these results, an M1 agonist seemed to facilitate the inhibition of old memory and the facilitation of reversal learning, which highlights the potential of M1 agonists to improve cognitive flexibility



Key takeaway:

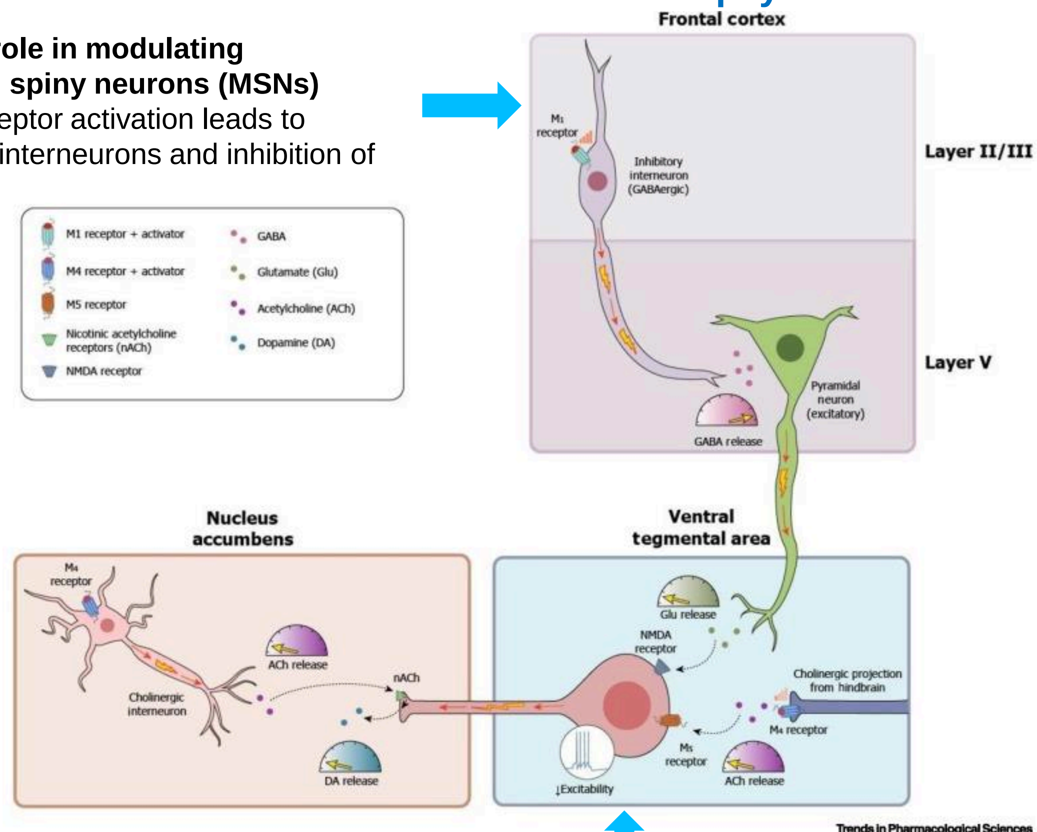
- **Mounting pre-clinical evidence supports the role of M1 agonists in enhancing cognitive flexibility, which further supports the development of M1 agonists or PAMs for targeting cognitive impairment in different CNS indications, including schizophrenia, ADP, and irritability in ASD**

MWM = Morris Water Maze; rMWM = reverse Morris Water Maze; WT = Wild type; PAM = Positive Allosteric Modulator
Source: Moran, Trends Pharmacol Sci, 2019; Melancon, Drug Discov Today, 2013; Xiong, Neuropharm, 2018; Yohn, Trends in Pharm Sci, 2022; Leerink Partners Research

Preclinical evidence suggests that M1 receptors mediate the antipsychotic effect in addition to their pro-cognitive effects; hence, they contribute to the robust antipsychotic efficacy along with M4 receptors

Activation of M1 receptors increases inhibition of pyramidal neurons in the frontal cortex, leading to decreased excitatory drive onto dopaminergic neurons in the midbrain and attenuation of psychosis

M1 receptors play a role in modulating dopaminergic medial spiny neurons (MSNs) excitability as M1 receptor activation leads to activation of inhibitory interneurons and inhibition of pyramidal neurons



Inhibition of pyramidal neurons reduces activation of dopaminergic neurons in the midbrain; and hence, reduction of psychosis

- **Further, preclinical evidence demonstrates the antipsychotic effect of M1 receptor activation:**
  - Global M1 KO mice exhibit a 'psychosis-like' phenotype, which suggests the potential antipsychotic efficacy of M1 receptor activation
  - And blocking of M1 receptors leads to hyperlocomotor activity in mice models
  - M1 PAMs reverse the excessive locomotor activity in a mouse model of NMDA-receptor mediated hypofunction phenotype

Key takeaways:

- **A growing body of evidence suggests that M1 receptors play a role in regulating neural circuits implicated in psychosis; it was suggested that the pharmacology of M1 agents determines their antipsychotic efficacy (e.g., activation of Gαq pathways leads to antipsychotic efficacy)**
- **The data readouts from M4 agents in 2024 suggested that the additional targeting of M1 receptors in M1/M4 agents may contribute to greater antipsychotic efficacy compared to M4 agents only; Cobenfy has set a high bar; thus, we believe that the clinical risk with ML-007C-MA in schizophrenia is lower given the similar MoA to Cobenfy (dual targeting of M1/M4 receptors)**

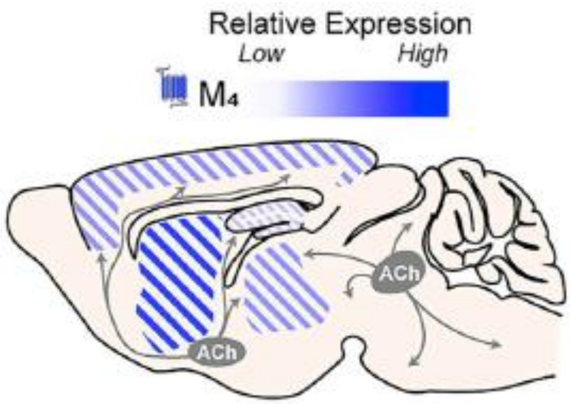
KO = Knockout

Source: Woolley, Eur J Pharm, 2009; Yohn, Trends in Pharm Sci, 2022; Gerber, Proc Natl Acad Sci USA, 2001; Grannan, ACS Chem Neurosci, 2016; Bradley, Nat Chem Biol, 2020; Yohn, Trends in Pharm Sci, 2022; Leerink Partners Research

M4 receptor expression in the brain in addition to strong preclinical evidence support the antipsychotic effect of M4 activators (either agonists/PAMs)

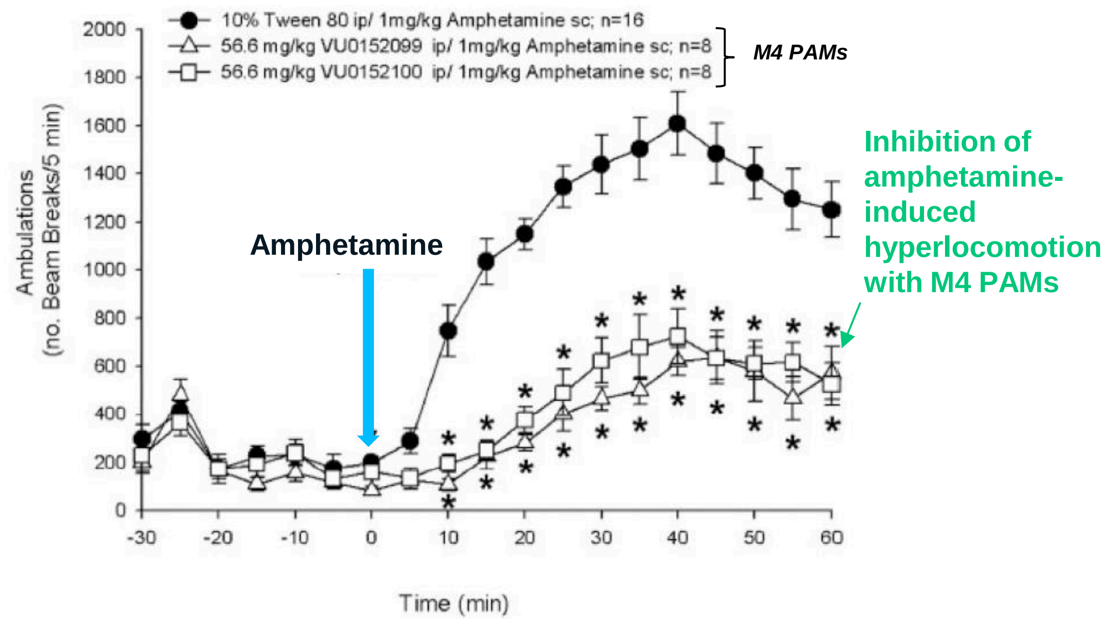
M4 receptor activators emerge as novel therapeutic agents for the treatment of psychosis in schizophrenia

High expression of M4 receptors in the striatum



- M4 receptors are highly expressed in the striatum, moderately expressed in the frontal cortex and the thalamus and poorly expressed in the hippocampus
- High concentration of **M4 receptors in the striatum implicates their role in psychosis mediated by aberrant dopaminergic neurotransmission in the mesolimbic pathway**

M4 PAMs inhibit the hyperlocomotion-induced by amphetamine



Amphetamine induces psychosis-like state/hyperlocomotion by inducing dopamine release in the mesolimbic pathway, mimicking the pathology of psychosis in schizophrenia

Key takeaway:

- **Substantial evidence points to the antipsychotic potential of M4 activators, either agonists or PAMs, that function without the blockade of D2 receptors in the mesolimbic pathway that result in unwanted side effects**

PAM = Positive Allosteric Modulator

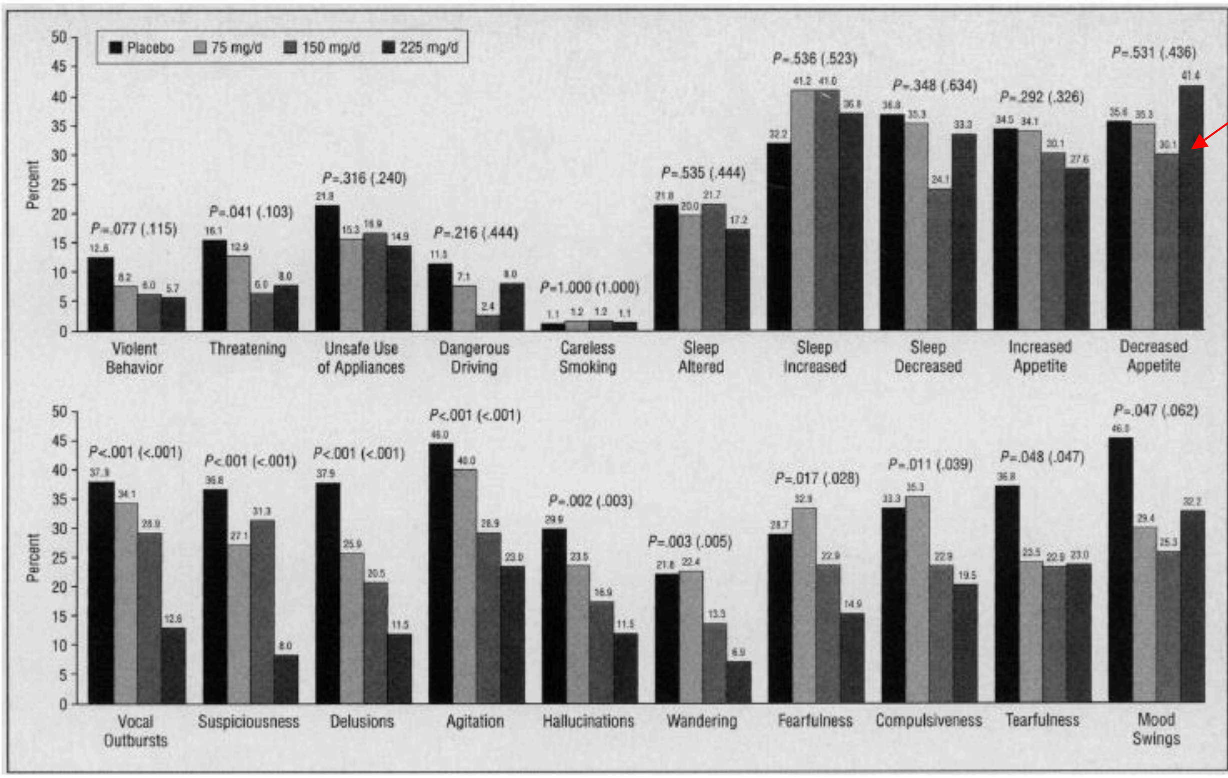
Source: Brady, J Pharmacol Exp Ther, 2008; Moran, Trends Pharmacol Sci, 2019; Leerink Partners Research

ML-007C-MA in Alzheimer's Disease Psychosis

The potential of ML-007C-MA in Alzheimer's disease psychosis is supported by prior data from xanomeline in Alzheimer's disease

Dose-dependent affects on ADSS with xanomeline

There was no effect on sleep, aggressive behavior or appetite



Fewer patients with xanomeline demonstrated worsening of their symptoms vs placebo

The dose-dependent effect on delusions, hallucinations, vocal outbursts and other symptoms suggests an effect of the drug on psychosis

Key takeaway:

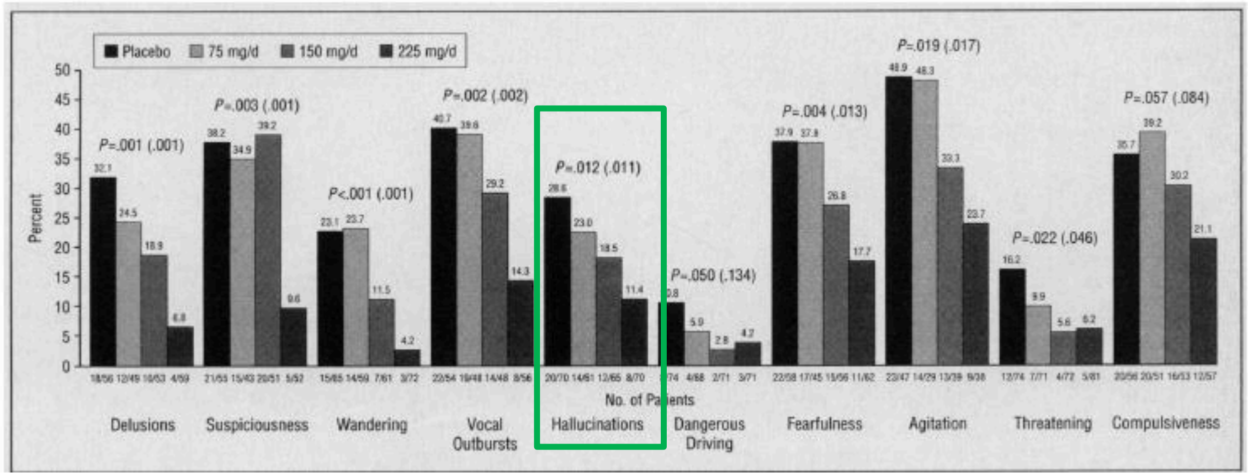
- **We believe the dose-dependent effect on delusions, hallucinations, vocal outbursts and other symptoms suggests an effect of the drug on psychosis, which is encouraging and bodes well for the Cobenfy ADEPT-2 study**

Sources: Company reports and presentations; Leerink Partners Research; Bodick et al., 1997,
ADSS = Alzheimer's disease composite score, ADAS-Cog = Alzheimer's Disease Assessment Scale- Cognitive Subscale; CIBIC = The Clinician Interview-Based Impression of Change

Xanomeline helped to prevent some of the disturbing behaviors, and patients remitted from their symptoms while being on xanomeline

Prevention

~89% of patients who had no hallucinations at baseline remained symptom free vs 71% on placebo with the highest dose of 225 mg/day

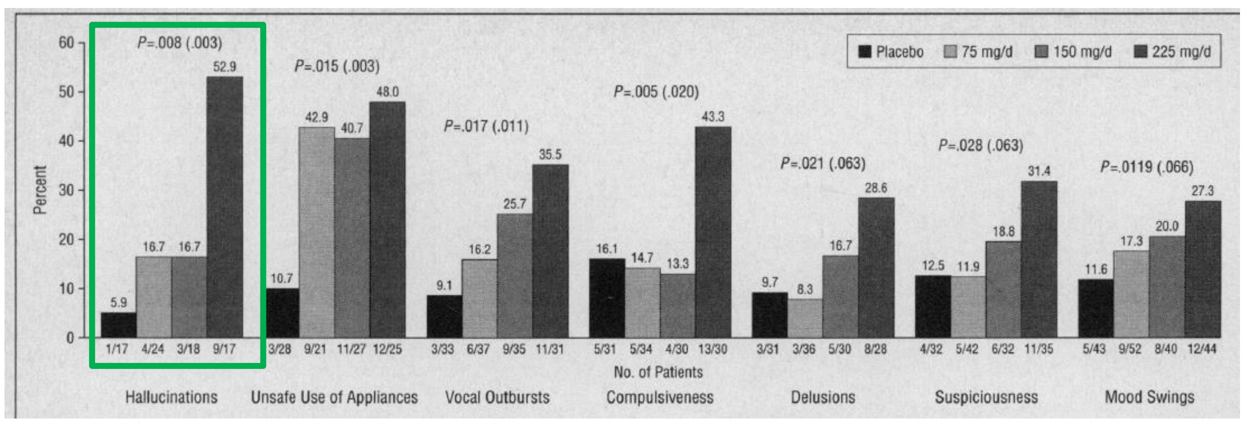


% refers to the proportion of patients with no symptoms at baseline

The majority of patients on xanomeline, who had no symptoms at baseline, remained symptoms free with therapy, and this effect was dose-dependent

Remission

53% of patients who experienced hallucinations at baseline fully remitted vs 6% on placebo with the highest dose of 225 mg/day



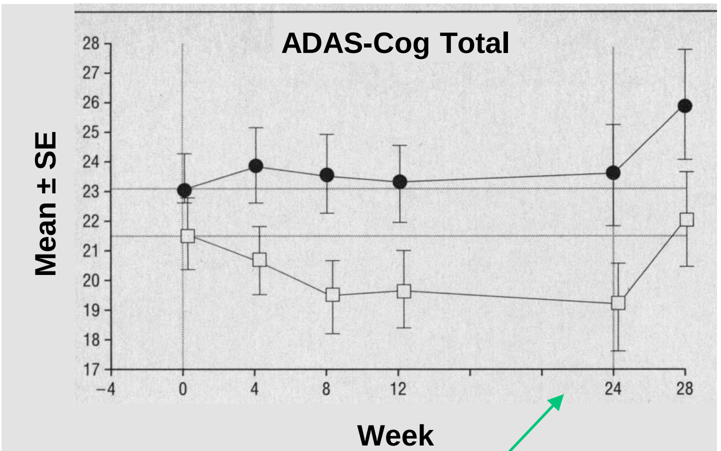
With higher dose, more patients remitted who had symptoms at baseline

Key takeaway:

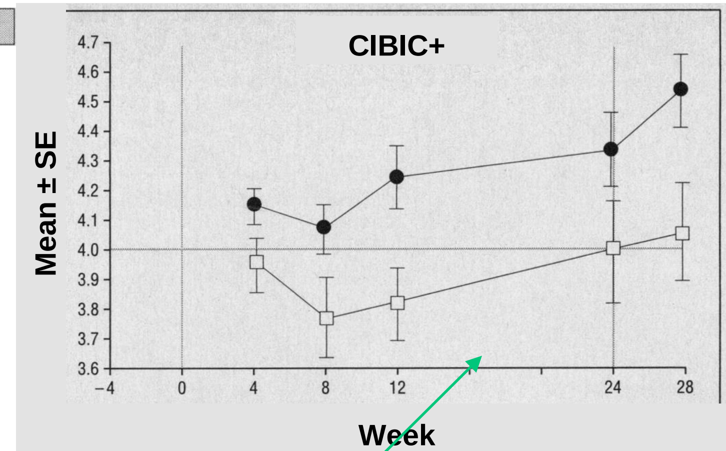
- **The positive effects in prevention and remission of xanomeline bode well for pending Cobenfy data in ADP**

Sources: Company reports and presentations; Leerink Partners Research ; *Bodick et al., 1997*
ADAS-Cog = Alzheimer's Disease Assessment Scale- Cognitive Subscale; CIBIC = The Clinician Interview-Based Impression of Change

Xanomeline (M1/M4 preferring agonist) also demonstrated positive effect on cognition after 6 months on therapy, contrary to the cognitive decline caused by use of antipsychotics



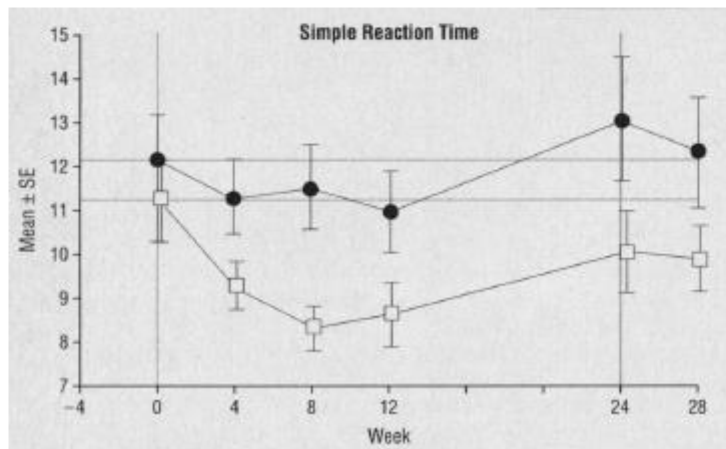
● Placebo □ 225 mg/d of Xanomeline Tartrate



A completer analysis showed significant reduction on ADAS-Cog11 ($p < 0.05$); xanomeline (high dose) specifically improved the following items: constructional praxis, orientation, spoken language ability and word-finding difficulty in spontaneous speech

However, the endpoint analysis was not significant (p=0.22)

Pairwise comparisons of the high dose arms with placebo indicated a stat sig drug effect (p=0.02)



Key takeaway:

- **These findings represent the first clinical evidence of the effect of activation of M1 and M4 receptors on improving cognition and behavioral symptoms, such as psychosis in AD, which is critical as this MoA offers a potential therapeutic solution to the elderly beyond the antipsychotics that are associated with mortality risk and cognitive impairment**
- **We look forward to the imminent data release from the Cobenfy ADEPT-2 trial in ADP expected in 4Q25, which should further validate this MoA in this indication**

Sources: Company reports and presentations; Leerink Partners Research ; Bodick et al., 1997
ADAS-Cog = Alzheimer's Disease Assessment Scale- Cognitive Subscale; CIBIC = The Clinician Interview-Based Impression of Change

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ML-007C-MA demonstrated a favorable safety and tolerability profile in the Phase 1 study in healthy elderly patients with BID dosing

	Placebo Combined N = 7 [2]	210/3 mg BID (with 2-7d titration) N = 11 [2]
Subjects with Any TEAE [1]	5 (71%)	8 (73%)
Mild	5 (71%)	8 (73%)
Moderate	0	2 (18%)
Severe	0	0
Most Common TEAEs (>2 Subjects)		
Hyperhidrosis	0	3 (27%)
Nausea	0	3 (27%)
Dizziness	1 (14%)	3 (27%)
Headache	0	4 (36%)
Tremor	0	3 (27%)

Importantly, there are no anti-cholinergic AEs

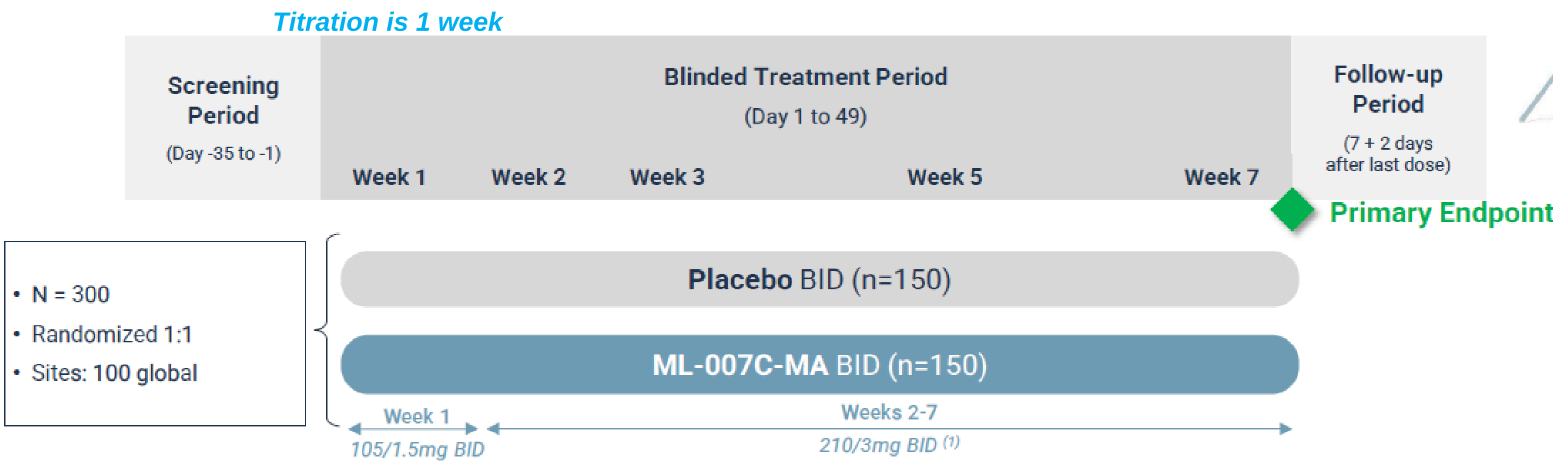
- For Cobenfy, the combination of the M1/M4 agonist to the PAC has to be modified to lower the concentration of PAC in the ratio, and TID dosing is used with a long titration period (5 weeks) to mitigate the AEs
- Also, Cobenfy is contraindicated in patients with urinary retention and untreated narrow-angle glaucoma; however, for ML-007C-MA, urinary retention probably won't be contraindicated given the lack of anti-cholinergic AEs and titration is shorter

Key takeaways:

- **We believe that the profile of ML-007C-MA in ADP is potentially improved over that of Cobenfy given the improved dosing and tolerability, which we believe should contribute to greater adoption among the ADP patients**

Sources: Company reports and publications; Leerink Partners Research

MapLight is running a Phase 2 trial in ADP with a parallel design with data expected in 2H27



VISTA

Study Population	• 55 to 90 years of age • Possible or probable Alzheimer's disease • Psychotic symptoms for >2 months • NPI-C H+D score of ≥6; and ≥2 on at least two items of either the H or D domain scores • CGI-S score ≥4	**Primary Endpoint**	• CFB to Week 7 on the NPI-C H+D domain scores
		Key Secondary Endpoints	• CFB to Week 7 on CGI-S score • CFB to Week 7 on NPI-C A+A domain score in participants with moderate agitation at Baseline

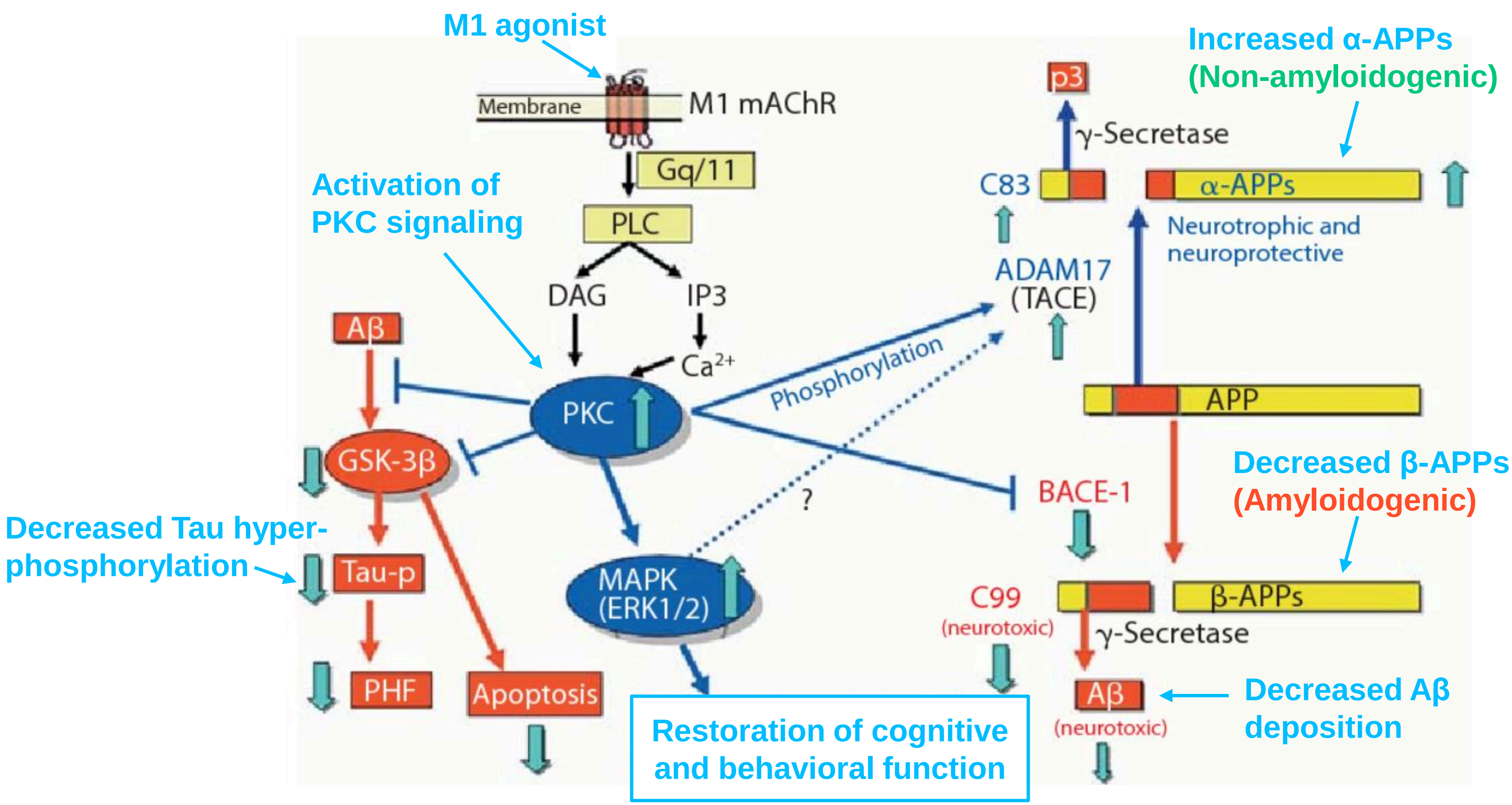
Key takeaways:

- **We look forward to the upcoming data from Cobenfy in ADP in 4Q25, which would help validate the mechanism in this significant indication and partially de-risk this opportunity for MapLight**

Sources: Company reports and publications; Leerink Partners Research

M4 agonists are believed to mediate the antipsychotic effects, and M1 agonists are believed to mediate the pro-cognitive effects in AD patients (similar to the effect in schizophrenia)

Additionally, M1 agonists could potentially reduce Aβ and Tau aggregation



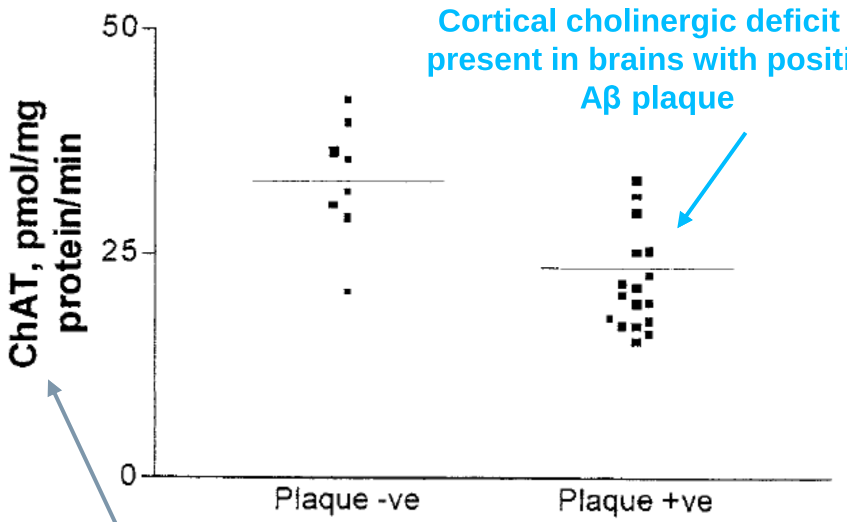
Key takeaway:

- **In addition to their role in enhancing cognition through potentiation of NMDA receptors (outlined HERE), M1 agonists could reduce the underlying neuropathology in AD**

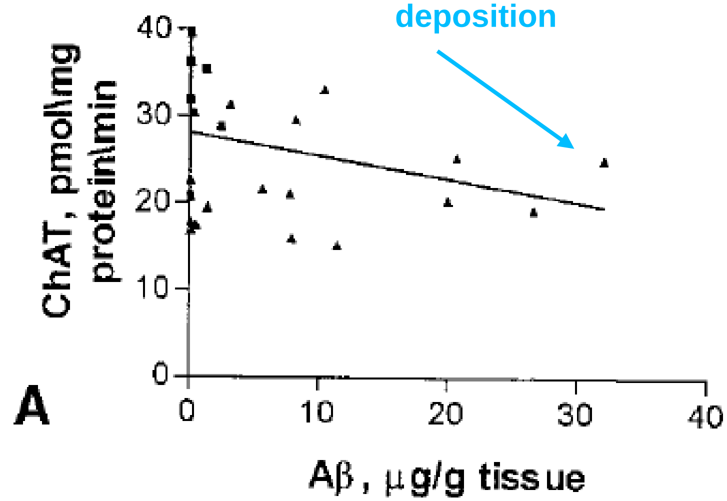
Sources: Company reports and presentations; Leerink Partners Research; Fisher, 2008
PKC = protein kinase C; APP = Amyloid precursor protein

Cholinergic denervation appears to be correlated with Aβ plaque deposition in early AD patients, suggesting that cholinergic deficits could be related to disease progression

Cortical cholinergic deficit is present in brains with positive Aβ plaque



Less cholinergic fiber density is associated with greater Aβ deposition



- **Choline Acetyltransferase (ChAT) is responsible for biosynthesis of acetylcholine and is used to label and measure density of cholinergic nerve fibers**
- Note that **loss of cholinergic neurons and impairment of presynaptic acetylcholine synthesis** in the cortex and the hippocampus has also been observed in AD

Key takeaway:

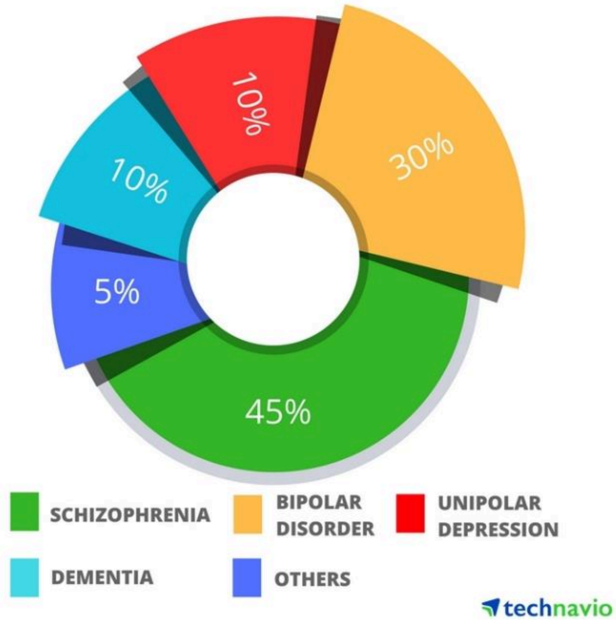
- **Increasing cholinergic neurotransmission with muscarinic agonists or positive allosteric modulators could play a role in reducing AD pathology or progression, including development of psychotic symptoms**

Sources: Company reports and presentations; Leerink Partners Research
AD = Alzheimer's disease

The high unmet need in the space of ADP

There is a lack of approved drugs for ADP, and off-label use of antipsychotics has resulted in poor benefit-risk profile in the elderly population

2016 Antipsychotics market share



*Credit: Businesswire.com and technavio**

- **The relative prevalence of psychosis varies according to the underlying dementia diagnosis**
  - The prevalence is 30-40% in AD , 75% in DLB , and 50% in PD dementia, while the prevalence is much lower in both vascular dementia and frontotemporal dementia with 15% and 10%, respectively
- **ADP is the most common type of dementia-related psychosis**
  - There are >6M AD patients in the US, representing 60-80% of all dementia patients
  - ~30-40% of AD patients (~1.8- 2.4 M) in the US experience psychosis with only ~50% of ADP patients currently diagnosed
- **The psychotic symptoms are the most clinically relevant neuropsychiatric symptoms (NPS) in AD patients**
  - Psychotic symptoms consist of hallucinations and delusions and are associated with high risk of hospitalization and caregiver distress
- **ADP is an area with high unmet need with no FDA approved drugs**
  - **~10% of atypical antipsychotics are used off-label in dementia and have severe side effects with the elderly,** such as cardiovascular events, EPS, orthostatic pressure, and excessive sedation due to the targeting of the dopaminergic pathways
  - **Atypical antipsychotics are associated with increased risk of death in the elderly** in addition to acceleration of cognitive decline

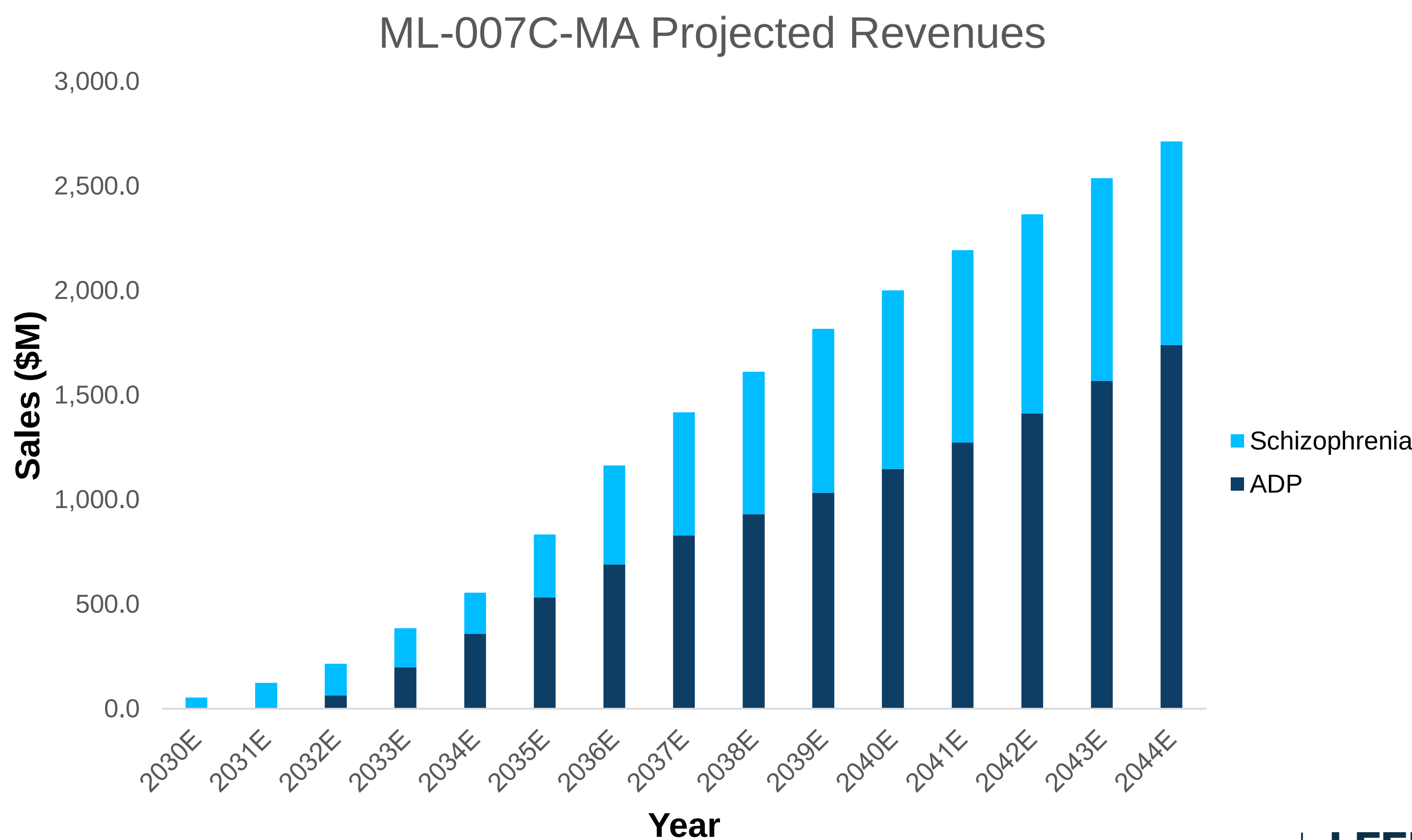
Key takeaway:

- **There is clearly a high unmet need, an opportunity large enough for multiple players**

Source: Ismail, Nat Rev Neur, 2022; Technavio and Businesswire.com; Company reports and presentation; Leerink Partners Research

ML-007C-MA – Sales Forecasts

- **Assumptions:**
  - **ADP patient population**: We assume 40% of ~6M patients with AD have psychosis and 30% are treated for an addressable population of ~725K
  - **Price per Rx:** We assume $25K/patient/year at launch with 5% annual price increases
  - **Gross-to-net (GTN):** We assume 35%
  - **LOE:** IP through 2045, assuming 5-year PTE



Source:Leerink Partners Research

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ML-004 in Autism Spectrum Disorder

ML-004 is an IR/ER formulation of the 5-HT1B/1D receptor agonist, which has the potential to increase sociability and reduce aggression through activating pro-social circuits in the nucleus accumbens

What's the hook?

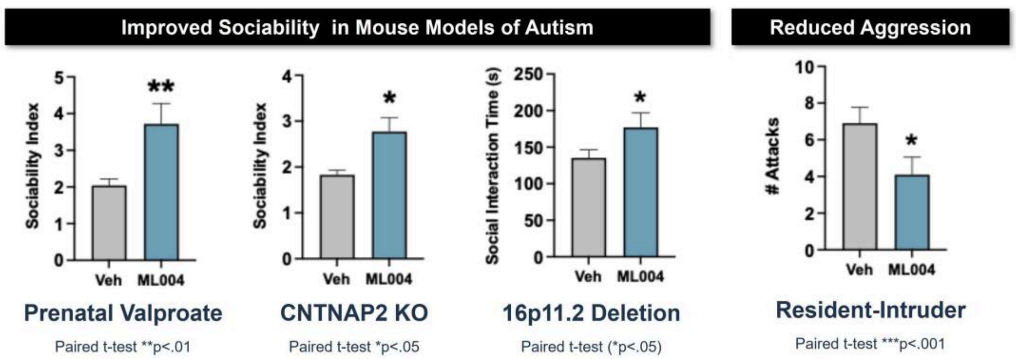
- The only two approved therapies in ASD are antipsychotics, which are associated with serious side effects, including weight gain and metabolic changes, EPS, and sedation
- Preclinical work demonstrated that 5-HT1B agonists increased sociability in ASD models and decreased aggression
- ML-004 is an IR/ER formulation of the 5-HT1B/1D agonist zolmitriptan, which has been commercially available for migraine headache since the 90s and has the potential to improve the core symptoms of ASD through the 5-HT1B pathway

MoA

- Zolmitriptan activates 5-HT1B/1D receptors located in the nucleus accumbens (NAc) to facilitate pro-social activity, and activating these circuits can improve sociability and reduce aggression

Data Highlights

ML-004 improved sociability and decreased aggression in several mouse models for ASD



- Published data has shown 5-HT1B agonism led to improved sociability in mice
- ML-004 was sufficient to improve social behavior in 4 different mouse models for ASD

Development status

- The Phase 2 IRIS study for social communication deficits in autism spectrum disorder is ongoing, and the company expects top-line results in 2H26

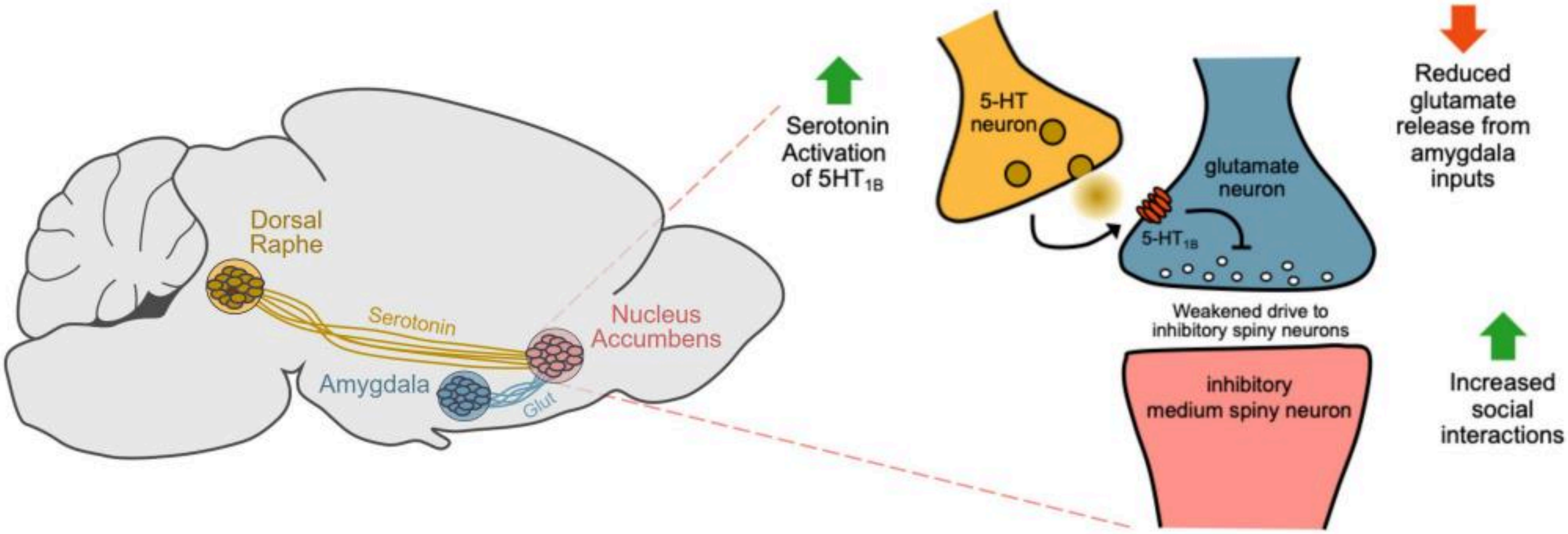
Patents

- MapLight owns pending US and foreign patent applications for the use of ML-004 for treating the symptoms of autism spectrum disorder, which expire in 2040 not including the potential for patent term extensions
- Additional IR/ER formulation patents for oral ML-004 are pending and expected to expire in 2042

Sources: Company reports and publications; Leerink Partners Research

It is hypothesized that sociability requires the activation of 5-HT1B receptors on excitatory glutamatergic amygdala inputs to the nucleus accumbens

Circuit diagram (mouse)



- It is hypothesized that sociability requires the activation of 5-HT1B receptors on excitatory glutamatergic amygdala inputs to the nucleus accumbens
- Reduced glutamatergic drive of postsynaptic nucleus accumbens neurons increases the reward associated with social interactions to promote sociability

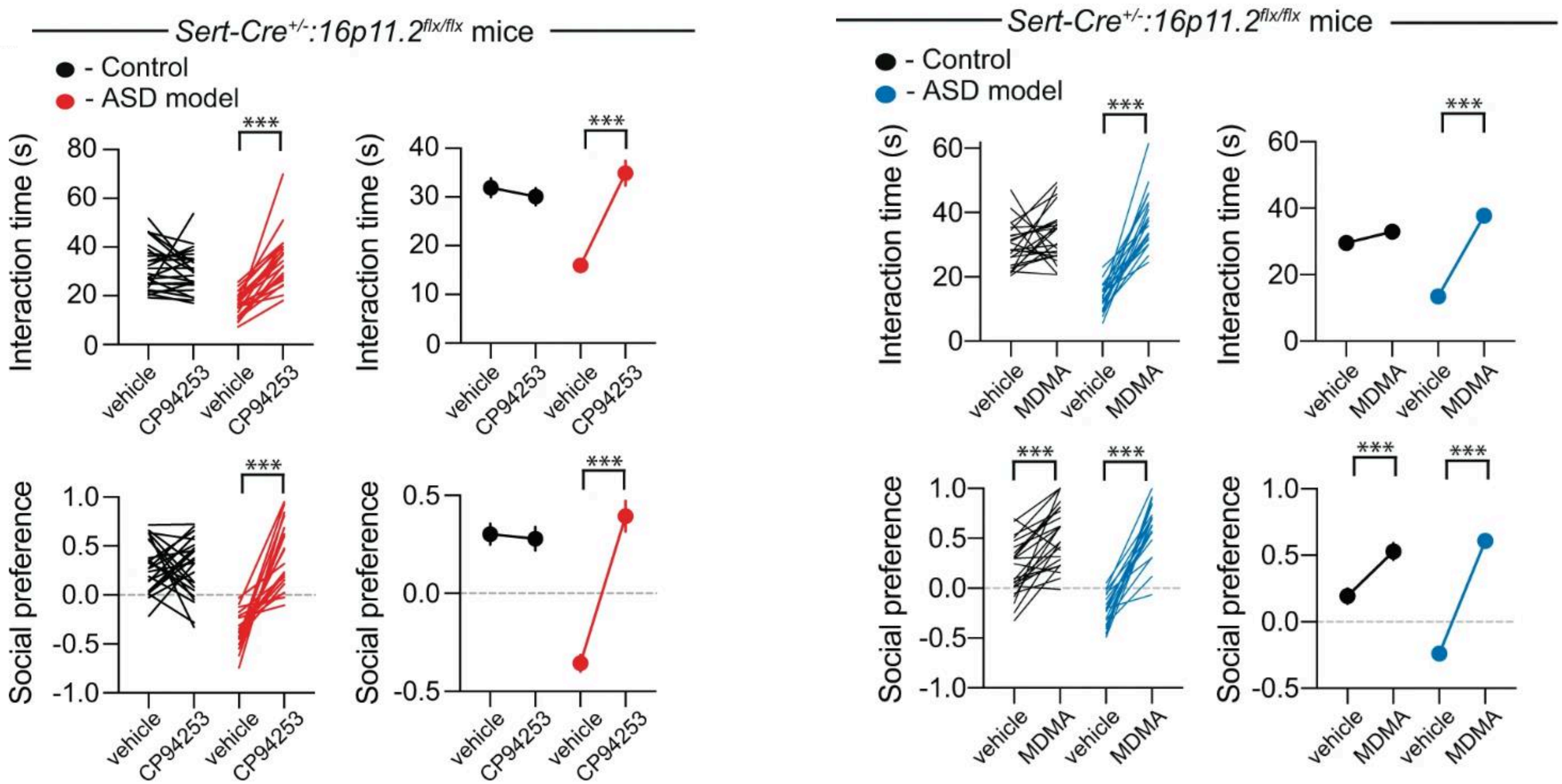
Key takeaway:

- **5-HT1B receptor activation is hypothesized to promote activity in the nucleus accumbens, leading to increased sociability**

Company reports and presentation; Leerink Partners Research

Preclinical data demonstrates a compelling effect on social behavior with 5HT1B agonism

5HT1B improves social behavior in an ASD model but not controls



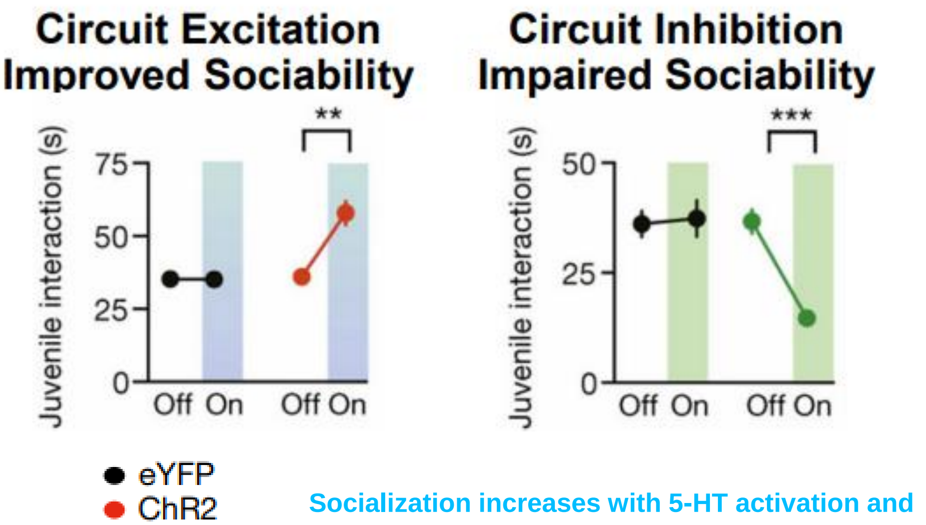
- An old laboratory molecule CP-94253 (5HT1B agonist) rescues social behavior deficits in the 16p11.2-deletion mouse model for autism but not controls, while MDMA improves social behavior in both controls and mutants
- This suggests that 5HT1B may target specific pathways that are disrupted in ASD circuits, while MDMA engages multiple circuits that may be unrelated to ASD
- Not shown: A study with data demonstrating the same effects across 4 distinct mouse models for ASD

Key takeaway:

- **5HT1B agonism is sufficient to rescue social deficits seen in multiple mouse models for ASD**

5-HT signaling increases sociability in juvenile mice, and this effect is suppressed with 5-HT1B antagonism

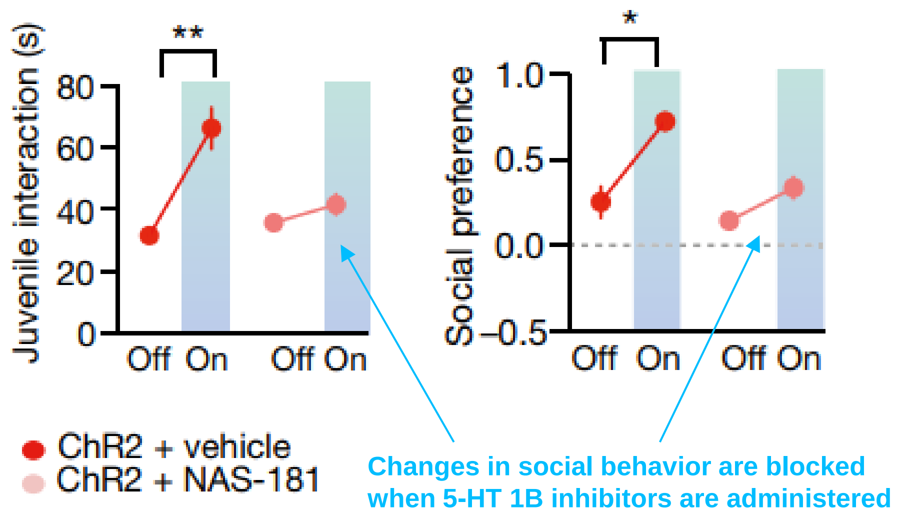
Mice are more social with 5-HT neuron activation



Socialization increases with 5-HT activation and suppressed with 5-HT inhibition

- Optogenetic activation of 5-HT neurons in the nucleus accumbens (ChR2, red) leads to increased social interactions in juvenile mice compared to controls (eYFP, black)
- Optogenetic suppression of 5-HT neurons in the nucleus accumbens (green) leads to decreased social interactions and in juvenile mice compared to controls (black)
- This indicates that serotonin release into the nucleus accumbens modulates social behavior in mice

5-HT1B inhibition blocks changes in social behavior



Changes in social behavior are blocked when 5-HT 1B inhibitors are administered

- The authors repeated the experiment showing increased social behavior with 5-HT neuron activation in the nucleus accumbens (ChR2 + vehicle, red)
- When the 5-HT1B inhibitor was infused into the nucleus accumbens before optogenetic stimulation, the social behavior was unchanged (ChR2 + NAS-181, pink)
- This suggests that serotonin-mediated sociability depends on the 5-HT1B receptor

Key takeaway:

- **Preclinical experiments demonstrate that the 5-HT1B receptor plays a role in social behavior**

Company reports and presentation; Leerink Partners Research, Walsh et al 2018

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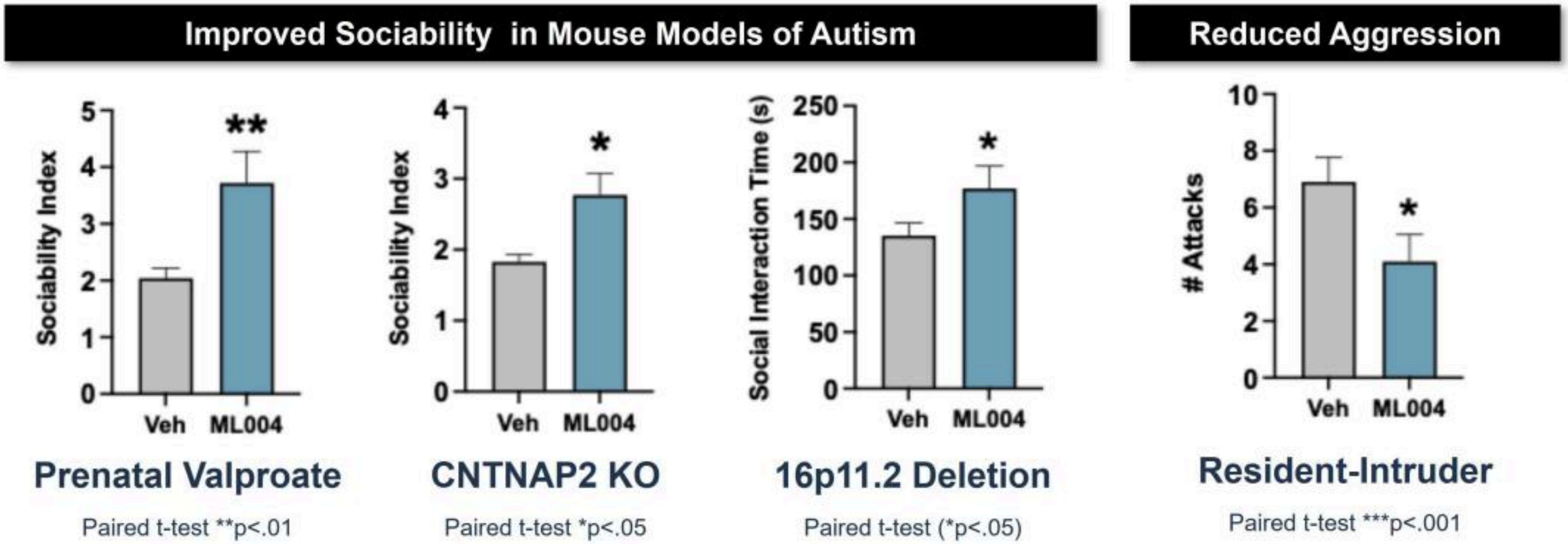
Preclinical studies with ML-004 demonstrated improvements in sociability and aggressive behaviors in multiple mouse models for ASD

Preclinical data with ML-004



- ML004 improved sociability in 3 distinct mouse models for autism
- ML004 reduced aggression in wild-type mice in an intruder-based assay

Key takeaway:

- **While preclinical models are imperfect, early evidence in mice demonstrate that ML-004 influences social behaviors, which suggests that ML-004 may lead to improvements in sociability in humans with ASD**

Company reports and presentation; Leerink Partners Research

MapLight completed 2 Phase 1 trials of ML-004 that supported safety and tolerability in addition to improved absorption compared to IR zolmitriptan

Phase 1 SAD trial

Design:

- Randomized, double-blind, placebo-controlled MAD PK trial testing high-dose IR zolmitriptan oral formulation in 40 healthy adults
- 5 ascending dose cohorts of 5mg, 10mg, 20mg, 30mg, 40 mg or placebo TID
- 2-4 day up-titration period, 7-day treatment period, 2-5 day down-titration period
- 40 adult healthy volunteers

Findings:

- The most common AEs were headache, dizziness, nausea, and hiccups
- No SAEs

Phase 1 bioavailability trial

Design:

- Open-label bioavailability PK Phase 1 trial evaluating ML-004 as a zolmitriptan bi-layer IR/ER gastroretentive 24 mg oral tablet formulation in 12 healthy adults under fasted and fed conditions
- 20mg zolmitriptan IR on Day 1, 24mg ML-004 fasted on Day 2, and 24mg ML-004 fed on Day 3

Findings:

- ML-004 led to prolonged absorption and reduced Cmax compared to IR zolmitriptan
- Based on these results, modeled data suggests that ML-004 48mg and 72 would result in target plasma exposure for 12-15 hours, respectively
- AEs were less common with ML-004 than IR zolmitriptan, and there were no discontinuations

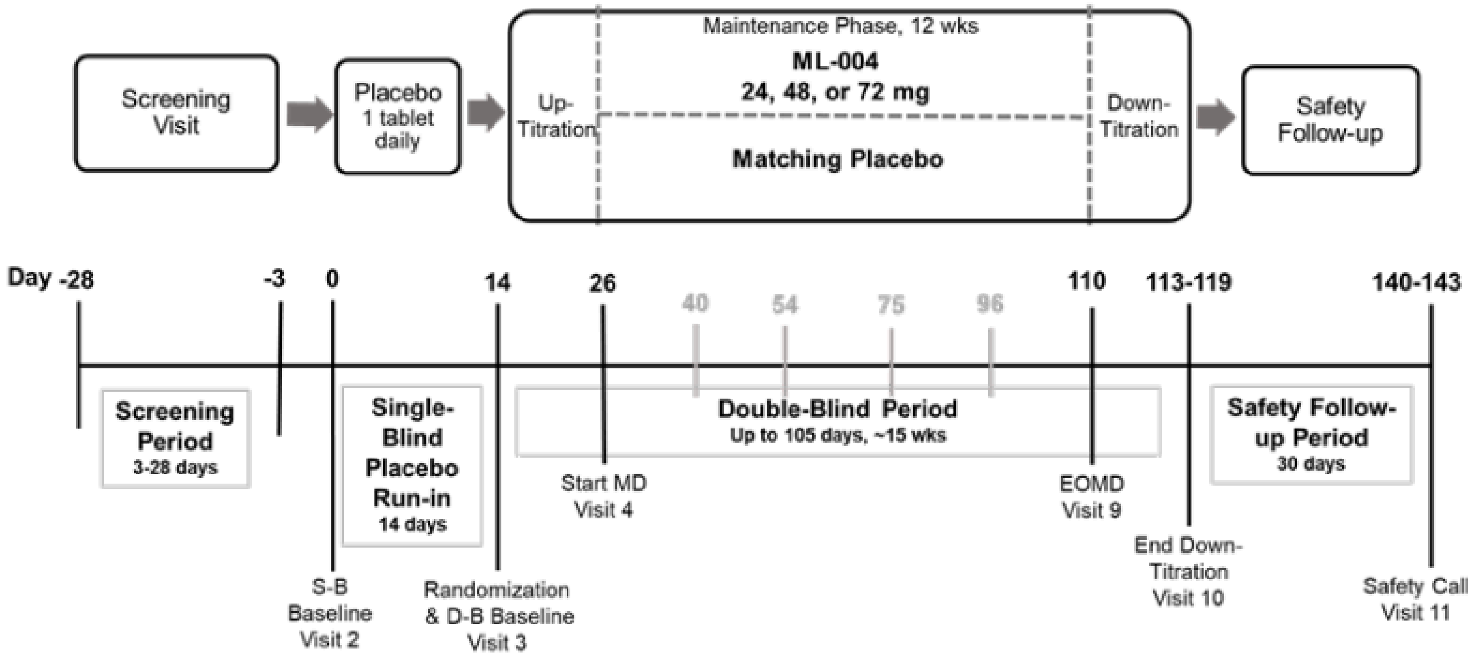
Key takeaway:

- **2 Phase 1 trials support safety and tolerability of ML-004**

Company reports and presentation; Leerink Partners Research

The Phase 2 IRIS trial will evaluate the efficacy of ML-004 for the improvement of social communication deficits in patients with ASD

Phase 2 trial design



Design:

- 12-week double blind period of 24mg, 48mg, 72 mg, or placebo
- Clinical sites in the US, Australia, and Canada and up to 150 participants are expected to be included

Primary endpoint:

- Autism behavior inventory – social communication domain

Key secondary measures:

- Clinical global impression of improvement
- Autism behavior inventory – clinician social communication domain
- Aberrant behavior checklist 2 – irritability

Key takeaway:

- **The Phase 2 IRIS trial of ML-004 will assess clinical endpoints for social communication and irritability in ASD**

Company reports and presentation; Leerink Partners Research

Key assets in development for autism spectrum disorder

Asset	Company	MoA	Indication	Available data	Next Catalyst	Timing
ML-004	MapLight	5-HT1B/1D agonist	Core ASD symptoms in adults	Phase 1 PK / bioavailability	Phase 2 topline data	1Q26
MM402	MindMed (MNMD, OP)	R(-)-MDMA	Core ASD symptoms in adults	None disclosed	Phase 2a initiation	4Q25
L1-79	Yamo (Private)	Tyrosine hydroxylase inhibitor	Core ASD symptoms in adolescents and young adults	Positive Phase 2 results	Phase 3 initiation	TBD
AST-001	Astrogen (Private)	SK channel agonist	Core ASD symptoms in children	Positive Phase 2 results	Phase 3 topline	Study completion 6/30/25
STP1	Stalicla (Private)	NKCC1 antagonist	Core ASD Phenotype 1	Phase 1b safety and PD biomarkers study	Phase 2 initiation	TBD
AJA001	DeFloria / Anja (Private)	Multi-cannabinoid botanical extract	Core ASD symptoms in children	Phase 1 safety study	Phase 2 initiation	4Q25
TTYP01	Shanghai Auzone (Private)	Oxidative stress pathways	ASD	-	Phase 2 study completion	2026
TB006	TrueBinding (Private)	Galectin-3	Core ASD symptoms in adults	-	Phase 2 study completion	2026
IAMA-6	IAMA (Private)	NKCC1	ASD	-	Phase 1 SAD/MAD	4Q25
Lumateperone	JNJ / Intracellular	Dopamine / serotonin receptor modulator	Irritability in ASD in children	-	Phase 3 study completion	2027
PRAX-100	Praxis (Not Rated)	SCN2A ASO	ASD	-	Candidate declaration	1H26
JAG201	Jaguar Gene Therapy (Not Rated)	AAV9-SHANK3	Genetic autism	-	Phase 1/2 study completion	2028

Sources: Company reports and presentations; BiomedTracker, Leerink Partners Research, ASO = antisense oligonucleotide

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MapLight Therapeutics
Management

Maplight Therapeutics Management Team

Christopher Kroeger, MD, MBA
CEO and board member

- More than 20 years of experience leading and building development-stage therapeutic and medical device companies
- Was previously the CEO of OvaScience, where he led a turnaround effort, resulting in a merger with Millendo Therapeutics
- Was a partner at the Aurora Funds

Erin Pennock, MD, PhD
CMO

- More than 13 years of experience in conducting and designing clinical trials
- Served as executive director of clinical development at Acadia and led the company's programs in the dementia/neurodegenerative space
- Held an academic physician-scientist position at the University of Virginia as a cognitive neurologist with a subspeciality interest in early-onset dementing illnesses

Vishwas Setia
CFO

- A decade of investment banking experience, advising biopharmaceutical companies and executing >$15B in equity and equity-linked financings and $100B in M&A and strategic transactions
- Was a managing director in the healthcare investment banking group at Bank of America Securities

James Lillie, PhD
CSO

- More than 25 years of experience driving preclinical research and development across diverse therapeutic areas
- Served as Chief Scientific Officer at OvaScience
- Was Vice President of In Vitro Biology at Sanofi Genzyme
- Supported the FDA's approval of Eliglustat

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Leerink Partners Financial Model

Maplight Therapeutics (MPLT): Income Statement

MapLight - Income Statement

($ Millions, except per-share data)

	2024	1Q25A	2Q25A	3Q25E	4Q25E	2025E	2026E	2027E	2028E	2029E	2030E
Total Revenues	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**53.7**
Consensus											
COGS	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6
Gross Profit	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	52.1
SG&A	14.4	3.5	4.1	4.7	5.6	17.9	19.2	20.1	21.1	23.8	179.9
R&D	68.5	22.0	24.6	43.8	49.5	140.0	200.0	225.0	230.0	215.0	180.0
Total Expenses	82.9	25.5	28.7	48.5	55.2	157.9	219.2	245.1	251.1	238.8	359.9
Operating Income	-82.9	-25.5	-28.7	-48.5	-55.2	-157.9	-219.2	-245.1	-251.1	-238.8	-307.8
Non-Operating Income	5.4	0.8	1.3	1.3	4.8	8.1	9.0	9.0	8.0	8.0	7.0
Pretax Income	-77.6	-24.8	-27.4	-47.2	-50.4	-149.8	-210.2	-236.1	-243.1	-230.8	-300.8
Taxes	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tax Rate							0.0%	0.0%	0.0%	0.0%	0.0%
Net Income	-77.4	-24.8	-27.4	-47.2	-50.4	-149.8	-210.2	-236.1	-243.1	-230.8	-300.8
Shares - Basic	26.1	26.2	26.2	26.2	37.9	29.1	48.9	57.5	57.8	65.0	71.6
EPS - Basic	**-$2.96**	**-$0.95**	**-$1.05**	**-$1.80**	**-$1.33**	**-$5.13**	**-$4.30**	**-$4.10**	**-$4.20**	**-$3.55**	**-$4.20**
Shares - FD											
EPS - Fully Diluted											

Margin Analysis	2024	1Q25A	2Q25A	3Q25E	4Q25E	2025E	2026E	2027E	2028E	2029E	2030E
Revenues											100.0%
COGS											3.0%
Gross Margin											97.0%
SG&A											
R&D											
Operating Margin											
Pretax Margin											
Net Margin											

% Change	2024	1Q25A	2Q25A	3Q25E	4Q25E	2025E	2026E	2027E	2028E	2029E	2030E
Revenues											
COGS											
Gross Profit											
SG&A	89.6%					24.1%	7.3%	4.7%	4.7%	13.0%	655.9%
R&D	37.9%					104.3%	42.9%	12.5%	2.2%	-6.5%	-16.3%
Operating Income	44.8%					90.4%	38.8%	11.8%	2.4%	-4.9%	28.9%
Pretax Income	39.3%					93.1%	40.3%	12.3%	2.9%	-5.0%	30.3%
Net Income	39.0%					93.5%	40.3%	12.3%	2.9%	-5.0%	30.3%

Sources: Companies reports and presentations; Leerink Partners Research

Maplight Therapeutics (MPLT): Revenues

MapLight - Revenues
($ Millions)

	1Q25A	2Q25A	3Q25E	4Q25E	2025E	2026E	2027E	2028E	2029E	2030E
Product Sales										
ML-007-MA - Schizophrenia										
US - Schizophrenia	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	53.7
% Growth Rate										
ML-007-MA - Alzheimer's Disease Psychosis										
US - ADP	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
% Growth Rate										
Total Sales - ML-007	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**53.7**
ML-004 - ASD										
US - ASD					0.0	0.0	0.0	0.0	0.0	0.0
Total Sales	**0.0**	**0.0**	**0.0**	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**	**53.7**
% Growth Rate										
Total Royalties & Milestones	**0.0**	**0.0**	**0.0**	**0.0**	0.0	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Total Revenues	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**53.7**
% Growth Rate										

Sources: Companies reports and presentations; Leerink Partners Research

Maplight Therapeutics (MPLT): Balance Sheet

MapLight- Balance Sheet

($ Millions)

	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
Assets								
Current Assets								
Cash and Cash Equivalents	79.8	38.3	405.5	445.1	411.9	181.0	155.6	60.7
Short-Term Investments	0.0	70.5	27.0	27.0	27.0	27.0	27.0	27.0
Accounts Receivable	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Inventory	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Prepaid Expenses & Other	3.8	5.8	12.7	12.7	12.7	12.7	12.7	12.7
Total Current Assets	**83.6**	**114.6**	**445.3**	**484.8**	**451.7**	**220.8**	**195.4**	**100.4**
Property and Equipment	1.0	1.2	1.0	1.0	1.0	1.0	1.0	1.0
Long-Term Investments	0.0	11.4	0.0	0.0	0.0	0.0	0.0	0.0
Restricted Cash	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Right of Use Assets	5.6	6.4	5.9	5.9	5.9	5.9	5.9	5.9
Other	1.0	3.2	3.8	3.8	3.8	3.8	3.8	3.8
Total Assets	**91.4**	**136.9**	**456.2**	**495.7**	**462.6**	**231.7**	**206.3**	**111.3**
Liabilities and Stockholder's Equity								
Current Liabilities								
Accounts Payable/Accrued Liabilities	3.3	1.9	2.1	2.1	2.1	2.1	2.1	2.1
Accrued Expenses	6.8	10.0	9.6	9.6	9.6	9.6	9.6	9.6
Lease Liability	0.5	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Deferred Grant Earnings	4.2	3.2	2.4	2.4	2.4	2.4	2.4	2.4
Total Current Liabilities	**14.7**	**15.9**	**15.0**	**15.0**	**15.0**	**15.0**	**15.0**	**15.0**
Other Liabilities	5.4	5.8	5.4	5.4	5.4	5.4	5.4	5.4
Total Liabilities	**20.1**	**21.7**	**20.4**	**20.4**	**20.4**	**20.4**	**20.4**	**20.4**
Stockholder's Equity								
Preferred Stock	188.9	308.8	0.0	0.0	0.0	0.0	0.0	0.0
Common Stock								
Additional Paid-in Capital	4.2	5.6	783.9	1,033.0	1,235.4	1,247.0	1,451.8	1,657.1
Accumulated Deficit	-121.8	-199.4	-350.1	-559.7	-795.2	-1,037.7	-1,268.0	-1,568.3
Accumulated Other Comprehensive Income/Loss	0.0	0.2	2.1	2.1	2.1	2.1	2.1	2.1
Total Stockholder's Equity	**71.3**	**115.2**	**435.8**	**475.4**	**442.2**	**211.3**	**185.9**	**91.0**
Total Liabilities and Stockholder's Equity	**91.4**	**136.9**	**456.2**	**495.7**	**462.6**	**231.7**	**206.3**	**111.3**

Sources: Companies reports and presentations; Leerink Partners Research

Maplight Therapeutics (MPLT): Cash Flow

MapLight - Cash Flow

($ Millions)

	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
Cash Flows from Operating Activities								
Net Loss	-55.7	-77.4	-149.8	-210.2	-236.1	-243.1	-230.8	-300.8
Depreciation	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Stock-Based Compensation Expense	1.0	1.1	9.0	10.2	11.4	12.6	13.9	14.4
Amortization		-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7
Non Cash Lease	0.5	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Loss from Reevaluation of Preferred Stock	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Loss from equity investment	-1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0
Loss on Disposal of Property and Equipment		0.1	0.0	0.0	0.0	0.0	0.0	0.0
Change in Operating Assets and Liabilities	2.0	-3.1	-8.3	0.0	0.0	0.0	0.0	0.0
Net Cash Used in Operating Activities	**-52.0**	**-78.7**	**-149.5**	**-200.3**	**-225.0**	**-230.7**	**-217.2**	**-286.8**
Cash Flows from Investing Activities								
Change in Marketable Securities	0.0	-80.0	54.9	0.0	0.0	0.0	0.0	0.0
Capital Expendititures	-0.5	-0.8	-0.1	-0.1	-0.1	-0.2	-0.2	-0.2
Other	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Net Cash Used in Investing Activities	**-0.5**	**-80.8**	**54.8**	**-0.1**	**-0.1**	**-0.2**	**-0.2**	**-0.2**
Cash Flows from Financing Activities								
Proceeds from Convertible Stock	104.6	120.0	192.0	0.0	0.0	0.0	0.0	0.0
Change in Common Stock, Net of Costs	0	0	269.5	240.0	192.0	0.0	192.0	192.0
Other	0.0	-1.9	0.0	0.0	0.0	0.0	0.0	0.0
Net Cash Used in Financing Activities	**104.6**	**118.1**	**461.5**	**240.0**	**192.0**	**0.0**	**192.0**	**192.0**
Cash/Cash Equivalents at Beginning of Period	27.9	80.0	38.7	405.5	445.1	411.9	181.0	155.6
Change in Cash	52.1	-41.4	366.9	39.6	-33.2	-230.9	-25.4	-95.0
Cash/Cash Equivalents at End of Period	**80.0**	**38.7**	**405.5**	**445.1**	**411.9**	**181.0**	**155.6**	**60.7**

Sources: Companies reports and presentations; Leerink Partners Research

Disclosures Appendix

Analyst Certification

I, Marc Goodman, certify that the views expressed in this report accurately reflect my views and that no part of my compensation was, is, or will be directly related to the specific recommendation or views contained in this report.

Distribution of Ratings/Investment Banking Services (IB) as of 09/30/25

Rating	Count	Percent	IB Serv./Past 12 Mos. Count	IB Serv./Past 12 Mos. Percent
BUY [OP]	221	75.4	111	50.2
HOLD [MP]	69	23.5	14	20.3
SELL [UP]	3	1.0	1	33.3

Explanation of Ratings

Outperform (Buy): We expect this stock to outperform its benchmark over the next 12 months.

Market Perform (Hold/Neutral): We expect this stock to perform in line with its benchmark over the next 12 months.

Underperform (Sell): We expect this stock to underperform its benchmark over the next 12 months.

The degree of outperformance or underperformance required to warrant an Outperform or an Underperform rating should be commensurate with the risk profile of the company.

For the purposes of these definitions the relevant benchmark for "Leerink Partners" branded healthcare and life sciences equity research will be the S&P 600® Health Care Index for issuers with a market capitalization of less than $2 billion and the S&P 500® Health Care Index for issuers with a market capitalization over $2 billion.

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The recommendation contained in this report was produced at November 21, 2025 0:02 A.M. EDT. and disseminated at November 21, 2025 6:00 A.M. EDT.

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